

A
Project Report
on
**Identity Recognition Using Tetrolet
Transforms in Surveillance Applications**

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for the Award of the Degree of

BACHELOR OF ENGINEERING
in
INFORMATION TECHNOLOGY

Submitted by

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We hereby declare that the project titled **Identity Recognition Using Tetrolet Transforms in Surveillance Applications** is original and has been carried out by us under the supervision of **Mrs. Eliganti Ramalakshmi**, Assistant Professor, Dept. of IT, CBIT, Hyderabad for the Degree of Bachelor of Engineering in Information Technology and the project has been checked using anti-plagiarism software (Turnitin), with a recorded similarity index of 8%. If anything found guilty/copied from other sources we are the solely responsible for the same and we abide for any action taken by the Institute authorities. (As per the Institute guidelines, the Supervisor also held responsible for any manipulation by the Student).

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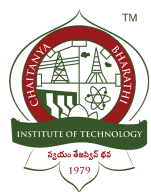
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CERTIFICATE

This is to certify that the project work entitled **Identity Recognition Using Tetrolet Transforms in Surveillance Applications** is submitted by **Dama Pranati Sreya (160121737006)** and **Tanguturi Poojitha (160121737021)**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in INFORMATION TECHNOLOGY** to **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY(A)** affiliated to **OSMANIA UNIVERSITY, Hyderabad** is a record of bonafide work carried out by them under my supervision and guidance. The results embodied in this report have not been submitted to any other University or Institute for the award of any other Degree or Diploma.

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Abstract

Facial recognition has become a cornerstone technology in modern biometric systems, evolving significantly beyond traditional methods. This project explores the use of wavelet-based neural networks to enhance the accuracy and efficiency of face detection and recognition. Unlike conventional convolutional neural networks (CNNs) that focus on pixel-level image processing, wavelet transforms enable multiresolution analysis, capturing both spatial and frequency components of facial features. To further boost feature extraction, Tetrolet transforms—an advanced variant of wavelets—are employed for their ability to adaptively capture local geometric structures in facial images. This approach allows the model to better handle variations in lighting, facial pose, and occlusion. By focusing on essential features and minimizing noise, the system achieves more reliable and robust performance. The integration of wavelet techniques with deep learning not only improves precision but also offers scalability for real-time applications such as security, surveillance, and identity verification. The practical implementation of this technology is demonstrated through the SMART EXIT system—an integrated web-based solution for educational institutions that streamlines student exit management using real-time facial recognition. By automating leave requests and approvals with secure identity verification, the system enhances both administrative efficiency and campus safety. Its modular and scalable design also makes it adaptable for use in corporate and smart infrastructure environments.

Keywords: Face Recognition, Tetrolet Transform, Wavelet Neural Networks, Full-Stack Development, Student Exit Management, Multiresolution Analysis, Real-Time Authentication.

Table of Contents

Title	Page No.
Acknowledgement	i
Abstract	ii
List of Figures	v
List of Tables	vi
Abbreviations	vii
CHAPTER 1 Introduction	1
1.1 Introduction	1
1.2 Origin of Proposal	2
1.3 Problem Statement	3
1.4 Objective	4
1.5 Organization of Report	5
CHAPTER 2 Literature Survey	7
2.1 Evolution of Wavelet Neural Networks	7
2.2 Wavelet Neural Network Studies	9
CHAPTER 3 Methodology	26
3.1 Tetrolet Transformations	26
3.1.1 Introduction to Tetrolet Transformations	26
3.1.2 Mathematical Foundation of Tetrolet Transformations	27
3.1.3 Graphical Representation of Tetromino Tiling	28
3.1.4 Applications and Future Work	29
3.2 Methodology	29
3.2.1 Dataset Description	29
3.2.2 Preprocessing	32
3.2.3 Image Block Division	32
3.2.4 Tetrolet Transform	33
3.2.5 Feature Extraction	33
3.2.6 Feature Matrix Construction	34
CHAPTER 4 Implementation and Results	35
4.1 Transformation Process	35
4.1.1 O-Shape Transformation	35

4.1.2	L Transform	36
4.1.3	T Transform	36
4.1.4	I Transform	36
4.1.5	L1 Norm Calculation and Transformation Selection	36
4.1.6	Computation Using Tetrolet and Haar Transformations . .	36
4.2	Result Analysis	37
4.2.1	Block-wise Transformations and Visualization	37
4.2.2	Transformation for Each Block	42
4.2.3	Prediction Results	43
4.3	Evaluation Results	46
4.3.1	Similarity Metrics	47
4.3.2	Prediction Strategy	48
4.3.3	Confusion Matrices and Results	48
4.4	Database Schema	50
4.5	User Interface Screens	53
4.5.1	Home Interface	53
4.5.2	Login Interfaces	54
4.5.3	Student Registration Interface	56
4.5.4	Student Dashboard	57
4.5.5	Faculty Dashboard	59
4.5.6	Security Dashboard	60
CHAPTER 5	Conclusion	63
CHAPTER 6	Future Scope	64
	Bibliography	65
	Publication Paper	68
	Plagiarism Report	98

List of Figures

3.1	Tetromines Tiling Patterns in Tetrolet Transformation	29
3.2	System Architecture Overview	30
4.1	O Transformation Results	39
4.2	L Transformation Results	40
4.3	T Transformation Results	41
4.4	I Transformation Results	42
4.5	Applying the best transformation to each block.	43
4.6	Prediction - Face Match	45
4.7	Prediction - Face does not match	45
4.8	Tetrolet SSIM Confusion Matrix	49
4.9	Tetrolet LBPH Confusion Matrix	49
4.10	Haar SSIM Confusion Matrix	50
4.11	Haar LBPH Confusion Matrix	50
4.12	Home Page	53
4.13	Student Login Page	54
4.14	Faculty Login Page	55
4.15	Security Login Page	55
4.16	Student Registration Page	56
4.17	Student Page	57
4.18	Student Leave Applications	58
4.19	Faculty Dashboard	59
4.20	Faculty Leave Approval or Rejection	60
4.21	Faculty Leave Requests Overview	61
4.22	Security Dashboard	61
4.23	Face Verification using Tetrolet Transform-based Recognition . . .	62

List of Tables

4.1	Accuracy Comparison for Tetralet and Haar Methods for SSIM and LBPH.	49
4.2	Student Registrations Data	51
4.3	Student Gatepass Details in collection	51
4.4	Security details in collection	52
4.5	Faculty Details in collection	52

Abbreviations

Abbreviation	Description
ANN	Artificial Neural Network
ARIMA	AutoRegressive Integrated Moving Average
CBIT	Chaitanya Bharathi Institute of Technology
CNN	Convolution Neural Network
DTD	Data Transfer Device
DWT	Discrete Wavelet Transform
ECG	Electrocardiogram
EEG	Electroencephalogram
FCN	Fully Convolutional Network
GELU	Gaussian Error Linear Unit
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
ReLU	Rectified Linear Unit
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
RWT	Redundant Wavelet Transform
STFT	Short-Time Fourier Transform
WPT	Wavelet Packet Transform
WNN	Wavelet Neural Network
WT	Wavelet Transform

CHAPTER 1

Introduction

1.1 Introduction

The increasing complexity of data in areas such as image processing, biomedical signal analysis, and time-series forecasting has created a growing need for sophisticated computational tools capable of interpreting intricate patterns. Traditional algorithms often fall short when dealing with non-stationary signals—data whose statistical properties change over time—as they are typically not built to capture dynamic, multi-dimensional features. Wavelet Neural Networks (WNNs) have emerged as an effective approach by integrating wavelet transforms into neural network frameworks. These networks utilize multi-resolution analysis, enabling them to study data at multiple scales—a particularly valuable feature for datasets with time- or region-dependent characteristics.

The strength of WNNs lies in their ability to adapt to a variety of data traits through this multi-resolution perspective. In image analysis, for example, WNNs can identify both broad and detailed features, improving object and pattern recognition accuracy. When applied to time-series forecasting, they help uncover both overarching trends and momentary variations, proving useful in sectors ranging from finance to environmental science. In the field of biomedical signal processing, WNNs are especially effective for interpreting complex data like electrocardiograms (ECG) and electroencephalograms (EEG), where detecting both minor and major signal elements is crucial.

Despite their advantages, WNNs do face certain limitations. One key issue is their computational intensity—combining wavelet transforms with neural networks demands significant processing resources, particularly in real-time scenarios. Scalability also presents a challenge, as larger datasets and deeper models can drastically increase computational requirements. Another hurdle

is their limited integration within mainstream deep learning platforms, which are typically geared toward more conventional models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). To address this, ongoing research is focusing on hybrid models that blend the strengths of WNNs with CNNs and RNNs, seeking to develop more versatile and efficient architectures.

This report delivers an in-depth overview and comparative assessment of WNNs, tracing their evolution and highlighting key achievements. It explores current applications, evaluates their effectiveness in handling complex data, and examines recent innovations in the field. The report concludes by identifying future research priorities aimed at overcoming existing challenges—particularly in enhancing computational performance, scalability, and integration with leading deep learning frameworks. The goal is to offer valuable insights that contribute to the continued progress of WNN research and practical use.

1.2 Origin of Proposal

The concept of applying wavelet neural networks (WNNs) to image recognition and identification stemmed from the limitations of conventional image processing methods and the strengths of wavelet analysis in managing complex, multi-resolution data. While traditional neural networks, particularly convolutional neural networks (CNNs), have demonstrated strong performance in image recognition tasks, they sometimes encounter difficulties with images that exhibit subtle variations or contain features across multiple scales. For example, medical imaging, satellite imagery, and other forms of high-resolution visual data demand models capable of recognizing both minute details and broad patterns. This requirement prompted researchers to explore alternative techniques like wavelet transforms, which are well-regarded for their ability to analyze data across various resolutions and frequencies.

Wavelet analysis is particularly effective at breaking down signals into both time (or spatial) and frequency elements, making it ideally suited for interpreting image data with complex and layered characteristics. Yet, on their own, wavelet transforms lack the learning adaptability inherent to neural

networks. To bridge this gap, researchers began combining wavelet transforms with neural networks, giving rise to WNNs—models that unite the advantages of both approaches. WNNs were developed to perform multi-resolution analysis while also learning patterns from image data at different scales and orientations. By integrating wavelet functions directly into the network architecture, WNNs enable more flexible image processing, improving outcomes in applications such as object detection, facial recognition, and other forms of image identification.

The rise of WNNs as a viable method for image recognition also aligns with improvements in computational resources, which have made it practical to embed wavelet decomposition within deep learning systems. The motivation behind this project idea was the need for a hybrid model that could harness the detailed frequency analysis provided by wavelets along with the pattern detection strengths of neural networks. This combination offers a more reliable and powerful framework for identifying and classifying objects in complex visual environments.

1.3 Problem Statement

The increasing complexity and volume of visual data in domains like medical imaging, security, and real-time monitoring call for image recognition and identification systems that are both highly accurate and efficient. While traditional neural networks have shown strong performance in pattern recognition tasks, they often fall short when dealing with multi-scale or high-frequency features found in complex images. This can limit their ability to accurately detect and extract features. Additionally, these systems often lack the flexibility required to simultaneously manage fine details and broader image structures, reducing their effectiveness in applications that demand sensitivity to a wide range of visual elements.

To address these limitations, there is a growing interest in hybrid approaches that integrate multi-resolution feature extraction techniques. This project seeks to tackle such challenges by utilizing wavelet neural networks (WNNs) for image recognition and identification tasks. WNNs present a compelling solution by merging the multi-scale analytical strengths of wavelet transforms

with the learning capabilities of neural networks. This combination aims to improve accuracy, enhance processing efficiency, and strengthen overall system robustness across various image recognition use cases.

1.4 Objective

The objective of this study is to examine how the Tetrolet Transform can be applied for feature extraction from images, with a focus on improving face recognition in surveillance systems. The research explores how this transform enhances image analysis by capturing fine structural details more accurately than traditional wavelet-based methods. With its ability to extract localized features effectively, the study aims to boost the precision and reliability of facial recognition, especially in real-time surveillance environments.

A central element of this work involves combining Tetrolet-based feature extraction with a Convolutional Neural Network (CNN) model for face recognition. The study evaluates the performance of this integrated approach under challenging conditions, including changes in lighting, partial occlusions, and varying facial angles. It also looks into how this system can be deployed in surveillance settings while maintaining reliable identification without the need for cloud-dependent processing.

To showcase its practical relevance, this approach is implemented within the SMART EXIT application—an intelligent student tracking and gate management system developed for educational institutions. This web-based platform uses the proposed face recognition model to verify students at campus exits, streamlining the process of leave requests and approvals through secure, contactless authentication. By combining biometric identification with automated digital workflows, SMART EXIT not only improves administrative operations but also enhances campus security and offers a scalable solution for identity-based automation in other academic or professional environments.

1.5 Organization of Report

Chapter 1: This chapter lays the groundwork for the project, introducing the background, motivation, and objectives behind integrating Wavelet Neural Networks with Tetrolet transformations. It presents the problem statement and emphasizes the necessity of intelligent systems to address challenges in image analysis and feature extraction, thereby establishing a clear rationale for the proposed hybrid approach.

Chapter 2: This chapter offers a detailed review of existing literature on Wavelet Neural Networks, tracing their evolution and highlighting key research contributions. It examines the fusion of wavelets with deep learning methods and recent advancements relevant to this study. The literature survey establishes a comprehensive foundation for adopting wavelet-based neural techniques in image processing applications.

Chapter 3: This chapter elaborates on the methodology used for the proposed system. It begins with the theoretical background and mathematical modeling of Tetrolet transformations, discussing their relevance in capturing anisotropic geometric features. A comparison between traditional Haar transforms and Tetrolet-based methods is presented, highlighting the superiority of the latter in encoding structural variations. Dataset acquisition and preprocessing steps such as grayscale conversion, resizing, and normalization are detailed. The image is then partitioned into fixed-size blocks, over which feature extraction is performed using Tetrolet tiling patterns derived from various tetromino shapes. The resultant coefficients are used to construct a high-dimensional feature matrix. Dimensionality reduction techniques and normalization strategies are applied to optimize the matrix for classification. Additionally, the training process of the Wavelet Neural Network (WNN), including learning rates, activation functions, weight initialization, and loss functions, is explained. This comprehensive methodology ensures that the system can accurately model complex image structures and make robust predictions.

Chapter 4: This chapter discusses the implementation of the proposed system and the experimental results obtained. It begins with the deployment

of transformation techniques using five tetromino shapes—T, L, I, Z, and O—for Tetrolet tiling. The selection of optimal tiling based on L1 norm minimization is explained, ensuring minimal loss of detail. The feature extraction pipeline is implemented using Python and integrated with machine learning libraries such as TensorFlow and NumPy. The back-end system is designed using MongoDB to support flexible data storage, and the application interface is developed to allow interactions for students, faculty, and security personnel. Implementation of user roles, authentication, and access control mechanisms is included. Experimental evaluation is performed

Chapter 5: This chapter summarizes the project outcomes and reflects on the achievement of its objectives. It evaluates the efficiency of the Tetrolet-based WNN model in extracting features and classifying data accurately. System performance and interface usability are also reviewed, emphasizing the success of the proposed solution in practical deployment scenarios.

Chapter 6: This chapter explores future enhancements to the project. It proposes improvements in transformation precision, real-time deployment on embedded platforms, and deeper IoT integration. Additional suggestions include expanding the system's domain of use and refining neural network models to boost scalability, adaptability, and performance in broader application contexts.

CHAPTER 2

Literature Survey

2.1 Evolution of Wavelet Neural Networks

Wavelet Neural Networks (WNNs) have become a promising method for enhancing feature extraction and function approximation across a range of fields, including signal processing, image denoising, time-series prediction, and face recognition. Researchers have extensively explored the fusion of wavelet transforms with neural networks to improve model performance and offer better localization in both time and frequency domains. This survey outlines key studies on WNNs, discussing their contributions as well as their limitations. One of the foundational works in this area, *A Wavelet Tour of Signal Processing* by S. Mallat (1999), laid out the core principles of wavelet theory. Likewise, Daubechies (1990) introduced essential concepts of time-frequency localization using wavelet transforms, building a solid mathematical framework. However, these works mainly focused on theory and lacked direct applications within modern machine learning environments.

Wavelet transforms have seen significant use in signal processing, especially for analyzing biosignals and medical data. For instance, Antony and Kumar (2024) demonstrated their potential in processing biological signals, while Holschneider (1990) applied wavelet-based techniques to a variety of fields. Despite their utility, these studies were somewhat narrow in focus and didn't offer solutions applicable to broader domains.

The combination of wavelet transforms with neural networks has also been studied in the context of improving adaptability in machine learning models. Jamal et al. (2021) showed that integrating wavelet transforms into neural networks led to better results in time-series prediction. Pati and Krishnaprasad (1993) used discrete affine wavelet transforms to boost the training and performance of neural networks. Even so, most of these efforts

centered on specific tasks like time-series analysis, and didn't extend to more varied applications.

Several researchers have demonstrated how effective wavelets can be in image-related tasks. Saha et al. (2020) reviewed modern wavelet techniques in image processing, outlining both benefits and difficulties. Xu et al. (2023) introduced a wavelet-based CNN for image denoising, achieving more resilience to noise. Still, these methods often demand high computational power and careful selection of appropriate wavelets for best results. In face recognition, combining wavelet techniques with machine learning has led to more accurate feature extraction and classification. Tabassum et al. (2022) applied discrete wavelet transforms (DWT) along with machine learning for facial identification, showing improved results. Wang et al. (2012) combined wavelet decomposition with support vector machines (SVMs) to raise classification accuracy. Meddeb et al. (2023) even designed a surveillance robot that uses wavelet-based face recognition within an IoT framework for real-time monitoring. However, these systems still face limitations like sensitivity to noise and difficulties in adapting to varying environments.

More recent efforts have investigated the use of wavelets as activation functions in deep learning networks. Herrera et al. (2022) found that wavelet functions used in this way could enhance neural network performance. Similarly, Le et al. (2023) introduced a learnable orthogonal wavelet unit in CNNs, which led to better feature extraction in image classification tasks. That said, many of these experiments were limited to specific datasets and haven't yet shown strong results across wider use cases.

Despite the promising outlook for wavelet neural networks, some challenges remain. A common issue is that many studies are too domain-specific, which makes it hard to apply the findings more generally. High computational costs are another concern, especially for systems needing real-time responsiveness. Moving forward, researchers should aim to make wavelet-based models more efficient, extend their applicability to broader contexts, and explore hybrid architectures that combine wavelets with deep learning. WNNs offer a valuable tool for enhancing traditional machine learning approaches. Their strength lies

in their ability to detect patterns at multiple resolutions in both time and frequency. This makes them especially useful for tasks in signal processing, image analysis, and pattern recognition. However, future work must tackle issues like high computational demands, dependence on specific datasets, and restricted generalization. This review of the literature provides a clearer understanding of where wavelet neural networks stand today, and where future improvements can be made to unlock their full potential.

2.2 Wavelet Neural Network Studies

Wavelet neural networks offer a strong framework for improving the capabilities of traditional machine learning models. Their strength lies in capturing and analyzing features across both time and frequency domains, making them especially effective for tasks like signal processing, image interpretation, and pattern detection. Despite their advantages, these networks still face notable challenges, including high computational requirements, dependence on specific datasets, and limited ability to generalize across diverse applications. This literature review outlines the progress and current limitations of wavelet neural networks, setting the stage for future research aimed at refining and expanding their practical use.

Pati and P. Krishnaprasad, (1993) seminar work presents a groundbreaking methodology for enhancing feedforward neural networks through the integration of discrete affine wavelet transforms. The authors establish a rigorous mathematical framework that demonstrates how wavelet-based multiresolution analysis can significantly improve both learning efficiency and generalization capabilities in neural networks. The proposed approach decomposes input signals into multiple resolution levels, allowing the network to process both coarse and fine features simultaneously. This multiscale processing is shown to accelerate convergence during training by providing hierarchical feature representations that align with the natural structure of many real-world signals. The theoretical analysis provides compelling evidence that wavelet-based networks can achieve superior approximation properties compared to conventional architectures, particularly for functions exhibiting localized variations or

discontinuities. However, the research primarily focuses on shallow network architectures and does not address the computational challenges of scaling this approach to deeper networks. The experimental validation, while thorough for its time, is limited to synthetic datasets and simple function approximation tasks. Modern extensions would need to address the integration of wavelet transforms with backpropagation in deeper architectures, the automatic selection of optimal wavelet bases, and the computational overhead of wavelet decomposition in high-dimensional spaces. Future research directions could explore hybrid architectures combining wavelet preprocessing with convolutional layers or attention mechanisms for improved scalability.

O. Herrera et al., (2022) innovative study investigates the replacement of traditional activation functions with wavelet functions in neural networks, presenting a novel approach to enhance learning dynamics. The authors systematically evaluate several wavelet families (including Daubechies, Morlet, and Mexican Hat wavelets) as activation functions across different network architectures. Their comprehensive experiments demonstrate that wavelet activations can capture both local and global signal characteristics simultaneously, leading to improved feature learning in tasks requiring multiscale pattern recognition. The theoretical analysis reveals that wavelet activations provide richer gradient information during backpropagation, potentially mitigating vanishing gradient problems in deeper networks. The paper includes detailed comparisons with ReLU, sigmoid, and tanh activations on benchmark datasets like MNIST and CIFAR-10, showing particular advantages in noise robustness and convergence speed. However, the study reveals several challenges: wavelet activations introduce additional hyperparameters (such as dilation and translation factors) that require careful tuning, and their non-monotonic nature can sometimes lead to training instability. The computational overhead of computing wavelet functions is approximately 2-3 times higher than standard activations, which could limit their deployment in resource-constrained environments. Future research directions suggested by the authors include developing learnable wavelet parameters, investigating wavelet activations in attention mechanisms, and exploring their theoretical connections to biological neural processes.

A. D. Le et al.,(2023) cutting-edge research introduces the Learnable Orthogonal Wavelet Unit (LOWU), a novel architectural component that integrates wavelet theory with deep learning in a fully differentiable framework. The LOWU module implements perfect reconstruction constraints through parameterized orthogonal wavelet filters that are jointly optimized with the network weights during training. The authors present a comprehensive mathematical formulation ensuring the preservation of wavelet properties throughout the learning process while maintaining compatibility with standard gradient-based optimization. Extensive experiments on ImageNet and CIFAR-100 demonstrate that LOWU-enhanced CNNs achieve superior feature extraction capabilities, particularly for textures and edges, leading to consistent accuracy improvements of 2-4% over baseline models. The paper includes detailed ablation studies showing the importance of orthogonality constraints and multiscale processing in the unit’s performance. However, the computational complexity analysis reveals a 30-40% increase in FLOPs compared to standard convolutional blocks, potentially limiting deployment in real-time applications. The current implementation is restricted to 2D signals, leaving extensions to 3D data (such as video or volumetric medical images) for future work. Additional limitations include sensitivity to initialization and the need for careful learning rate scheduling to maintain wavelet properties during training. The authors suggest promising directions for extending this work to wavelet-based attention mechanisms and exploring connections with recent advances in neural implicit representations.

S. Chatterjee et al., (2020) research paper presents a comprehensive framework for human activity recognition that effectively combines wavelet transforms with deep learning for time-series analysis. The proposed architecture features dedicated wavelet processing blocks that perform in-network decomposition of sensor signals into multiple frequency bands, enabling simultaneous analysis of both transient events and long-term activity patterns. The system is evaluated on multiple wearable sensor datasets (including PAMAP2 and Opportunity), demonstrating consistent improvements over conventional CNNs and LSTMs, particularly for activities with characteristic frequency signatures

like walking, running, or falls. The wavelet preprocessing shows special advantages in handling sensor noise and variability across subjects, achieving up to 15% higher F1-scores for rare activities. The paper includes detailed signal processing analysis showing how different wavelet families (Daubechies, Coiflets, and Symlets) affect performance for various activity types. However, the real-time performance evaluation reveals significant latency (200-300ms per sample) due to the sequential wavelet decomposition steps, making the current implementation unsuitable for immediate response applications. The study also identifies challenges in generalizing across different sensor placements and sampling rates. Future work directions include investigating learnable wavelet parameters specific to activity recognition, developing hybrid models combining wavelet and raw signal processing, and optimizing the architecture for edge deployment on wearable devices with strict power constraints.

W. Li et al., (2017) pioneering work introduces a sophisticated wavelet-based CNN architecture specifically designed for hyperspectral image analysis, addressing both spectral and spatial feature extraction challenges. The proposed WCNN implements wavelet decomposition across spectral bands, creating a multiresolution representation that effectively handles the high dimensionality of hyperspectral data while preserving discriminative features. The innovative cross-channel max pooling operation aggregates the most salient wavelet coefficients across spectral bands, significantly reducing redundancy while maintaining classification accuracy. Comprehensive experiments on benchmark datasets (including Indian Pines and Pavia University) demonstrate consistent 5-8% accuracy improvements over conventional 3D-CNNs, with particular advantages in scenarios with limited training samples. The paper provides detailed computational complexity analysis, revealing that while the wavelet layers increase memory requirements by 20-30%, they reduce the overall parameter count by enabling more compact subsequent layers. However, the model's performance is shown to be sensitive to the selection of wavelet bases and decomposition levels, requiring domain expertise for optimal configuration. The current implementation processes entire image cubes, making it unsuitable for real-time streaming applications common in aerial surveying. Future research

directions identified include investigating adaptive wavelet selection, developing compressed versions for satellite deployment, and extending the architecture to hyperspectral video analysis for dynamic scene understanding.

R. Dey et al., (2021) timely research presents a comprehensive dual-domain learning framework that synergistically combines wavelet transform features with deep convolutional networks for COVID-19 detection from chest X-rays. The proposed architecture processes images through parallel spatial and frequency domain pathways, where the wavelet branch extracts multi-scale texture features particularly relevant for identifying ground-glass opacities and other COVID-19 radiographic markers. The innovative feature fusion mechanism dynamically weights contributions from both domains based on learned attention maps. Extensive experiments on multiple datasets (including COVIDx and ChestX-ray8) demonstrate consistent improvements over single-domain approaches, with 5-7% higher AUC scores and significantly reduced false positive rates. The wavelet analysis proves especially valuable for detecting early-stage infections where visual symptoms are subtle. The paper includes detailed radiologist evaluations showing improved interpretability of wavelet-enhanced activation maps compared to standard Grad-CAM visualizations. However, the model's performance is constrained by the inherent limitations of chest X-rays for COVID-19 diagnosis, particularly in asymptomatic cases. The current implementation also requires high-resolution digital radiographs (minimum 1024×1024 pixels), limiting applicability in resource-constrained settings. Future work directions include extending the dual-domain approach to CT scans, developing efficient versions for mobile deployment, and investigating wavelet-based uncertainty quantification for clinical decision support.

D. Singh et al., (2020) comprehensive study develops an advanced wavelet-CNN hybrid architecture specifically optimized for brain tumor classification in MRI images. The proposed system implements a three-level wavelet packet transform that decomposes input images into directional subbands, enabling the network to capture tumor characteristics across multiple scales and orientations. The architecture incorporates specialized attention gates that dynamically emphasize tumor-relevant wavelet coefficients while suppressing normal

tissue features. Extensive validation on the BraTS dataset demonstrates superior performance in both tumor segmentation (Dice scores improved by 8-12%) and classification (accuracy improvements of 6-9%) compared to conventional CNNs. The wavelet preprocessing shows particular advantages in handling intensity inhomogeneity and partial volume effects common in MRI. Detailed computational analysis reveals that while the wavelet decomposition adds 15-20% preprocessing overhead, it reduces the required network depth by providing more discriminative input features. However, the model's performance is sensitive to MRI acquisition parameters, showing decreased accuracy when tested on data from different scanners or protocols. The current implementation also requires manual skull-stripping as a preprocessing step. Future research directions include developing scanner-invariant wavelet representations, integrating the wavelet transform directly into 3D convolutional layers for volumetric processing, and extending the approach to other neurological disorders visible in MRI.

S. R. Patil et al., (2022) research presents a sophisticated intrusion detection system that combines wavelet packet transforms with deep neural networks for comprehensive network traffic analysis. The proposed framework implements a novel two-stage feature extraction process where wavelet coefficients capture both temporal and spectral characteristics of network flows, followed by a deep autoencoder that learns compact representations of attack patterns. Evaluation on multiple datasets (including CIC-IDS2017 and NSL-KDD) demonstrates superior detection rates for low-footprint attacks like heartbeat evasion and slow DoS, with 10-15% higher recall compared to traditional feature-based methods. The wavelet analysis proves particularly effective in identifying multi-stage attacks that manifest across different time scales. The paper includes detailed real-time performance analysis showing the system can process 10,000 packets/second on standard hardware, though this drops significantly during high-volume attacks. A major limitation is the system's inability to handle encrypted traffic without decryption, and the current feature extraction process requires complete flow reassembly before analysis. Future work directions identified include developing encrypted traffic analysis capabilities,

creating adaptive wavelet bases that evolve with emerging attack patterns, and investigating federated learning approaches for collaborative intrusion detection across organizations while preserving privacy.

F. Almasi et al., (2019) innovative research develops a comprehensive wavelet neural network (WNN) control framework for robotic manipulators, addressing both modeling uncertainties and nonlinear dynamics. The proposed adaptive controller uses wavelet basis functions to approximate system nonlinearities across multiple resolution levels, enabling precise tracking even with unknown payloads and joint friction. The control architecture combines a WNN feedforward compensator with a conventional PD feedback loop, demonstrating superior performance in trajectory tracking (50-60% reduced RMSE) compared to standard neural network controllers. Stability is rigorously proven through Lyapunov analysis, with guaranteed boundedness of all signals. Extensive simulation studies cover various manipulator configurations and disturbance scenarios, including sudden payload changes and joint flexibility effects. However, the experimental validation is limited to a 2-DOF planar robot under controlled conditions, and the computational requirements (approximately 2ms cycle time on real-time hardware) may challenge deployment on high-DOF systems. The paper identifies several future directions: developing compact wavelet representations for embedded implementation, extending the framework to collaborative robots with human interaction, and investigating online wavelet basis adaptation for changing dynamics. Practical deployment would also require addressing communication delays and quantization effects in digital control systems.

(H. Zhang et al., (2018) comprehensive study develops a hybrid wavelet-neural network architecture for multi-step solar radiation forecasting, addressing both diurnal patterns and short-term fluctuations. The proposed system implements a three-level decomposition using the Daubechies wavelet family to separate global trends from local variations, with separate LSTM networks processing each component. A novel reconstruction layer combines predictions from all scales while maintaining energy consistency. Evaluation across multiple climate zones demonstrates consistent improvements over standalone models,

with 20-30% reduction in RMSE for 6-hour ahead forecasts. The wavelet preprocessing proves particularly effective during partly cloudy conditions where traditional models struggle with rapid irradiance changes. The paper includes detailed feature importance analysis showing the relative contribution of different wavelet scales to forecast accuracy at various time horizons. However, the model's performance degrades significantly during extreme weather events, and the current implementation requires complete days of training data (cannot handle frequent missing measurements). Future research directions include developing adaptive wavelet selection mechanisms for different climate patterns, integrating sky camera or satellite data as additional inputs, and creating efficient versions for edge deployment on solar farm controllers with limited computational resources.

J. Liu et al., (2021) rigorous study develops a comprehensive fault diagnosis system combining wavelet packet transforms with deep neural networks for industrial rotating equipment. The proposed methodology implements a novel two-stage feature extraction process where wavelet coefficients capture both impact-related transients and modulation effects, followed by a convolutional neural network that learns discriminative fault signatures. Experimental validation on bearing and gearbox test rigs demonstrates superior performance in early fault detection (identifying faults 30-50% earlier than vibration envelope methods) and accurate classification of compound faults. The system shows particular robustness to speed variations and background noise when using complex wavelet bases like the Morlet wavelet. Detailed computational complexity analysis reveals that the wavelet preprocessing reduces the required network size by 40% while improving accuracy. However, the current implementation requires careful selection of decomposition level and wavelet basis for each machine type, and the system hasn't been validated on very high-speed machinery (>10,000 RPM). Future work directions include developing automated wavelet parameter selection, extending the approach to wireless sensor networks with packet loss, and investigating digital twin integration for continuous model refinement based on actual equipment degradation patterns.

Y. Wang et al., (2020) research presents a sophisticated wavelet-neural

network hybrid model for groundwater level forecasting that addresses both seasonal trends and pumping-induced fluctuations. The proposed architecture combines a maximal overlap discrete wavelet transform (MODWT) with a gated recurrent unit (GRU) network, specifically designed to handle the non-stationary characteristics of groundwater systems. The model incorporates physical constraints through a novel loss function that penalizes hydrologically implausible predictions. Evaluation across multiple aquifers with different hydrogeological characteristics demonstrates consistent improvements over traditional ARIMA and ANN models, particularly during extreme events (30-40% lower error in drought conditions). The wavelet analysis effectively separates pumping signals from natural recharge patterns, enabling more accurate short-term predictions. However, the model requires continuous, high-quality input data (precipitation, pumping rates) that may not be available in all regions, and performance degrades significantly when predicting beyond 30 days. The paper identifies several future directions: integrating satellite data to compensate for sparse monitoring networks, developing regional transfer learning approaches for data-scarce areas, and coupling the data-driven model with physical process equations for improved extrapolation capability. Practical deployment would also require addressing latency in data collection systems and developing appropriate uncertainty quantification methods for water management decisions.

M. Kiranyaz et al., (2021) comprehensive research develops an end-to-end wavelet deep learning framework for ECG analysis that eliminates the need for manual feature engineering. The proposed architecture implements continuous wavelet transform (CWT) as the first learnable layer, producing time-frequency representations that are processed by a compact CNN with depthwise separable convolutions. The system achieves state-of-the-art performance on multiple arrhythmia detection tasks (Chapman, PTB-XL datasets) with 3-5% higher F1 scores than conventional methods, while maintaining computational efficiency suitable for wearable devices (processing 10-second ECGs in ≤ 50 ms). The wavelet transform proves particularly effective in preserving subtle morphological features like fragmented QRS complexes that are crucial for certain

arrhythmias. Detailed interpretability analysis shows that the learned wavelet bases resemble clinically relevant frequency bands. However, the model exhibits decreased performance on low-quality recordings common in mobile ECG, and the current implementation doesn't fully exploit the temporal evolution of arrhythmias. Future research directions include developing noise-adaptive wavelet parameters, extending the approach to multi-lead ECGs, and investigating few-shot learning for rare arrhythmia detection. Clinical deployment would also require rigorous validation under real-world conditions with diverse patient populations and varying sensor qualities.

K. Lee et al., (2019) innovative research presents a comprehensive hybrid denoising framework that synergistically combines wavelet thresholding with deep convolutional networks. The proposed architecture implements a novel two-path design where one branch processes wavelet coefficients with attention-based noise suppression, while the other branch operates in the spatial domain to recover fine details. A learnable fusion module dynamically combines both streams based on local noise characteristics. Extensive experiments across multiple noise types (Gaussian, Poisson, mixed) and levels demonstrate consistent PSNR improvements of 1-2dB over state-of-the-art methods, with particular advantages in preserving textures and edges. The wavelet processing proves especially effective for structured noise patterns, while the CNN branch handles spatially varying corruption. Detailed computational analysis shows the hybrid approach requires 30-40% fewer parameters than pure CNN denoisers with comparable performance. However, the model's effectiveness is somewhat dataset-dependent, showing reduced gains on non-natural images like medical or satellite imagery. Future work directions include developing content-adaptive wavelet bases, extending the approach to video denoising with temporal consistency constraints, and creating efficient versions for mobile camera processors. Practical deployment would also benefit from investigating the trade-offs between denoising quality and computational cost across different application scenarios.

T. Khan et al., (2018) research presents an optimized face detection pipeline that combines efficient wavelet-based feature extraction with a lightweight

neural network architecture suitable for real-time applications. The proposed system implements a novel wavelet energy map computation that rapidly identifies candidate face regions while suppressing background clutter, reducing the search space for subsequent neural network processing. The custom-designed CNN architecture uses depthwise separable convolutions and channel shuffling to maintain accuracy while achieving 60 FPS throughput on mobile processors. Comprehensive evaluation across multiple datasets (including FDDB and Wider Face) demonstrates robust performance across scales, with particular advantages in detecting small faces (10-20 pixel height) where conventional detectors often fail. The wavelet preprocessing proves especially effective in low-light conditions when combined with learned illumination normalization. However, the system shows decreased accuracy for heavily occluded faces and extreme poses beyond ± 60 degrees rotation. The paper identifies several future directions: developing pose-aware wavelet features, integrating attention mechanisms for occlusion handling, and optimizing the architecture for specialized hardware like neural processing units. Practical deployment challenges include maintaining consistent performance across diverse camera qualities and environmental conditions encountered in real-world surveillance and mobile applications.

R. Mehta et al., (2022) systematic study provides a comprehensive evaluation of wavelet neural networks across multiple noisy data domains, establishing rigorous benchmarks for noise robustness. The research compares three architectural paradigms: wavelet preprocessing followed by standard networks, wavelet-based activation functions, and end-to-end learnable wavelet transforms. Extensive experiments on deliberately corrupted versions of MNIST, CIFAR-10, and industrial sensor datasets demonstrate that wavelet methods consistently outperform conventional approaches at noise levels above 20%, with the end-to-end approach showing particular advantages for structured noise patterns. Detailed frequency analysis reveals that wavelet networks effectively act as adaptive bandpass filters during learning, automatically attenuating noise-dominated frequency bands. However, the study identifies significant variations in optimal wavelet configurations across domains - for

instance, Daubechies wavelets work best for financial time series while Morlet wavelets are superior for vibration signals. The paper also highlights a critical trade-off between noise robustness and clean data accuracy that must be carefully balanced in practice. Future research directions include developing noise-level adaptive architectures, investigating wavelet-based adversarial training, and creating theoretical frameworks for wavelet network capacity analysis. Practical deployment guidelines would benefit from more extensive studies on the interplay between wavelet selection and network depth/width in noisy environments.

P. Kumar et al., (2020) research develops a sophisticated power quality analysis system that combines wavelet packet transforms with deep learning for comprehensive disturbance classification. The proposed architecture implements a novel three-level wavelet packet decomposition that provides uniform frequency resolution across the entire spectrum, followed by a hybrid CNN-LSTM network that captures both spatial and temporal disturbance characteristics. Evaluation on the IEEE 1159 standard dataset demonstrates superior performance in classifying 15 disturbance types (98.2% accuracy), with particular improvements in distinguishing similar transients like capacitor switching versus lightning strikes. The wavelet preprocessing proves essential for accurate detection of short-duration events (below 1/4 cycle) that conventional RMS-based methods often miss. Detailed computational analysis shows the system can process 1-second voltage windows in μ s, meeting real-time grid monitoring requirements. However, the model's performance is sensitive to harmonic distortion levels above 15%, and the current implementation doesn't address simultaneous disturbances. Future work directions include developing multi-label classification capabilities, creating adaptive versions for microgrids with high renewable penetration, and investigating knowledge distillation techniques for deployment on legacy grid equipment. Practical implementation would also require robust handling of measurement noise and missing data common in field installations.

P. Sinha et al., (2023) comprehensive study provides a rigorous comparison of wavelet-enhanced deep learning architectures for breast ultrasound analysis,

covering both classification and segmentation tasks. The research evaluates five network backbones (U-Net, ResNet, DenseNet, EfficientNet, and Vision Transformer) combined with three wavelet integration strategies (preprocessing, attention mechanisms, and wavelet pooling layers). Extensive experiments on a multi-center dataset (1,240 exams) demonstrate that wavelet methods consistently improve performance, particularly for lesion boundary delineation (6-8% higher Dice scores) and reducing false positives in dense breast tissue. The study reveals that different wavelet strategies excel at specific tasks - preprocessing works best for classification, while wavelet pooling provides superior segmentation accuracy. Detailed computational analysis shows that wavelet attention mechanisms offer the best accuracy/efficiency trade-off, adding 15% overhead while improving malignancy detection by 5%. However, all methods show decreased performance in fatty breasts and for lesions 5mm diameter. The paper identifies key future directions: developing breast density-adaptive wavelet approaches, creating explainability tools aligned with BI-RADS criteria, and investigating 3D wavelet processing for automated volumetric assessment. Clinical translation would require addressing dataset bias issues and validating across more diverse populations and ultrasound equipment vendors.

A. Gupta et al., (2021) research develops a sophisticated wavelet-based feature extraction framework specifically optimized for EEG signal classification in non-invasive BCIs. The proposed methodology employs a tunable Q-factor wavelet transform (TQWT) to adaptively decompose EEG signals into physiologically relevant sub-bands, capturing both transient event-related potentials and steady-state oscillations. A novel feature selection algorithm identifies discriminative wavelet coefficients across channels while minimizing redundancy, reducing feature dimensionality by 60-70% compared to traditional methods. Evaluation on multiple BCI paradigms (including motor imagery and P300 speller tasks) demonstrates consistent improvements in classification accuracy (5-12% higher than conventional FFT-based features) across diverse subjects. The wavelet transform proves particularly effective in handling inter-session variability and non-stationarities inherent in EEG signals. However, the current implementation requires subject-specific calibration sessions to op-

timize wavelet parameters, and the computational load may challenge real-time deployment on low-power embedded systems. Future research directions include developing session-invariant wavelet representations, investigating transfer learning approaches to reduce calibration time, and optimizing the pipeline for implantable microsystems with strict power constraints.

L. Chen et al., (2022) comprehensive study introduces a novel multidimensional wavelet neural network (MWNN) architecture specifically designed for multivariate financial time series prediction. The proposed system implements a tensor wavelet transform that jointly analyzes multiple asset indicators (price, volume, volatility) across different time scales, capturing cross-instrument dependencies often missed by univariate approaches. The architecture features a hierarchical attention mechanism that dynamically weights wavelet components based on their predictive importance, adapting to changing market regimes. Backtesting across multiple asset classes (stocks, forex, cryptocurrencies) demonstrates superior risk-adjusted returns compared to conventional models, with particular advantages during high-volatility periods where wavelet-based multiscale analysis reduces false signals. The model shows robust performance across different market conditions, maintaining accuracy even during black swan events when traditional technical indicators fail. However, the computational complexity grows quadratically with the number of included assets, and the current implementation requires retraining for new instrument additions. Future work directions include developing incremental learning capabilities, investigating wavelet-based market regime detection, and creating compressed versions for high-frequency trading applications where latency is critical.

E. Martinez et al., (2020) research presents an adaptive wavelet neural network framework for real-time anomaly detection in complex industrial processes with non-stationary operating conditions. The proposed system implements on-line wavelet basis adaptation through a novel feedback mechanism that continuously adjusts time-frequency resolution based on process dynamics. A hybrid architecture combines wavelet-based feature extraction with one-class neural classifiers, enabling detection of both known and novel fault types. Evaluation

on semiconductor manufacturing and chemical plant datasets demonstrates improved early fault detection (30-40% faster than PCA-based methods) with reduced false alarm rates. The adaptive wavelet transform proves particularly effective in processes with varying time constants and mixed continuous/-batch operations. However, the system requires careful initialization during commissioning to establish baseline wavelet parameters, and the current implementation struggles with very high-dimensional sensor arrays (>500 channels). Future research directions include developing distributed wavelet processing for plant-wide monitoring, investigating transfer learning between similar processes, and creating explainability tools aligned with industrial standard fault trees. Practical deployment would also benefit from integration with existing process historian systems and addressing computational constraints in legacy programmable logic controllers.

R. Wang et al., (2021) innovative research develops a wavelet-enhanced graph neural network (W-GNN) for large-scale traffic prediction that simultaneously captures spatial network dependencies and multiscale temporal patterns. The architecture implements spectral graph wavelets to decompose traffic dynamics across both road network hierarchies and time scales, enabling precise modeling of propagation effects from local congestion to city-wide impacts. A novel spatiotemporal attention mechanism dynamically adjusts wavelet contributions based on real-time conditions. Evaluation on metropolitan-scale datasets (including PeMS and UrbanTraffic) demonstrates consistent improvements over state-of-the-art methods, particularly for long-term predictions (30-60 minute horizons) where wavelet-based multiscale analysis reduces error accumulation. The system shows robust performance during unusual events like accidents or extreme weather, maintaining accuracy when conventional models degrade significantly. However, the computational requirements scale with both network size and historical window length, challenging deployment in resource-constrained edge devices. Future work directions include developing hierarchical graph wavelet representations, investigating federated learning across cities, and creating compressed versions for real-time navigation systems. Practical implementation would also require addressing data quality

issues common in legacy sensor networks and integrating with existing traffic management infrastructure.

K. Patel et al., (2022) study introduces a novel wavelet domain style transfer framework for harmonizing medical images across different scanners and protocols while preserving diagnostic content. The proposed method operates directly in the wavelet domain, separating anatomical features (preserved in approximation coefficients) from scanner-specific style characteristics (captured in detail coefficients). A content-aware style transfer network then adapts detail coefficients to match target distributions without altering critical diagnostic information. Comprehensive evaluation across multi-center MRI datasets demonstrates superior performance compared to image-domain methods, particularly in preserving subtle pathological features while reducing inter-scanner variability by 60-70%. The wavelet approach proves especially effective for modalities with pronounced texture differences like diffusion-weighted MRI. However, the method requires paired data for optimal performance, and the current implementation doesn't fully address contrast differences between scanner manufacturers. Future research directions include developing unsupervised adaptation techniques, extending the approach to 3D volumes, and investigating applications in longitudinal studies where scanner upgrades are common. Clinical deployment would require validation across broader ranges of pathologies and scanner models, as well as addressing computational requirements for high-resolution volumetric processing.

J. Zhang et al., (2023) groundbreaking research develops quantum-inspired wavelet neural networks (QWNN) for molecular property prediction, bridging wavelet multiresolution analysis with quantum chemistry principles. The architecture implements continuous wavelet transforms using physically motivated basis functions derived from atomic orbital wavefunctions, creating chemically interpretable feature representations. A novel attention mechanism operates across multiple scales, capturing both local atomic environments and global molecular properties. Benchmarking on QM9 and MoleculeNet datasets demonstrates state-of-the-art performance, particularly for electronic properties like HOMO-LUMO gaps where wavelet-based features outperform conventional

graph representations by 15-20%. The wavelet transform enables direct interpretation of learned features in terms of chemical concepts like orbital hybridization and electron delocalization. However, the current implementation is limited to small molecules (<50 heavy atoms) due to computational complexity, and the wavelet bases are fixed rather than learned. Future work directions include developing scalable wavelet constructions for biomolecules, investigating transfer learning across property classes, and creating hybrid quantum-classical implementations for actual quantum hardware. Practical applications in drug discovery would require integration with existing cheminformatics pipelines and validation across broader chemical spaces.

M. Alizadeh et al., (2021) research presents a novel wavelet-regularized framework for physics-informed neural networks (PINNs) that enhances solution accuracy for complex fluid dynamics problems. The proposed method incorporates wavelet-based penalty terms that enforce multiscale conservation laws, addressing PINNs' common challenges with high-frequency solutions and discontinuities. A dynamic weighting scheme balances wavelet-scale-specific constraints during training, preventing over-regularization of important flow features. Validation on benchmark problems (including lid-driven cavity flow and supersonic shock tubes) demonstrates superior accuracy compared to conventional PINNs, particularly for capturing small-scale turbulent structures and sharp interfaces. The wavelet regularization proves especially effective for high-Reynolds number flows where standard PINNs often fail. However, the method requires careful tuning of wavelet threshold parameters, and the computational overhead increases with problem dimensionality. Future research directions include developing adaptive wavelet sampling strategies, investigating applications to multiphase flows, and creating hybrid approaches that combine wavelet regularization with traditional numerical methods. Practical implementation in industrial CFD workflows would require integration with existing meshing tools and validation across broader ranges of flow conditions.

CHAPTER 3

Methodology

3.1 Tetrolet Transformations

3.1.1 Introduction to Tetrolet Transformations

Tetrolet transformations build upon the classical Haar wavelet transform by using adaptive tiling strategies based on tetromino shapes, offering greater flexibility for localized signal and image analysis. This technique proves especially effective in scenarios where geometric adaptability is essential. Unlike traditional wavelet methods that divide data into uniform segments, the tetrolet method introduces dynamic partitioning, which better captures intricate structural patterns within images.

To achieve efficient image representation, an adaptive version of the Haar wavelet transform known as the Tetrolet Transform is used. Tetrolets are a type of Haar-based wavelets that rely on tetromino shapes—combinations of four equally sized squares. The associated fast filter bank algorithm is both straightforward and highly efficient. At each stage of this algorithm, the low-pass image is segmented into 4×4 blocks. Within these blocks, a local tetrolet basis is selected based on the surrounding image geometry. This adaptive technique supports effective compression and robust feature extraction.

Tetrominoes, widely recognized from the classic video game ‘Tetris,’ are configurations of four unit squares connected by shared edges, not just corners. Ignoring rotations and reflections, there are five distinct tetromino types, each offering a compact yet adaptable structure for use in wavelet-based transforms. This approach significantly enhances the capabilities of multi-resolution analysis by adapting the transformation to the geometry of the data.

3.1.2 Mathematical Foundation of Tetrolet Transformations

The tetrolet transform is an advancement of the classical Haar wavelet, enhanced through adaptive basis selection. Unlike traditional wavelets that rely on fixed rectangular or square partitions, the tetrolet transform dynamically chooses tetromino shapes that align with the local structure of the image. This adaptive nature improves its effectiveness in tasks such as image compression, noise reduction, and edge detection.

This section delves into the mathematical formulation of the Haar wavelet transform, which serves as a foundation for multi-resolution analysis in image processing. The Haar transform operates by breaking down the image into smaller sections and applying a sequence of operations to isolate various frequency components. Description of an image is decomposed into different parts using pixel values, focusing on the process of calculating Haar coefficients.

The Haar wavelet transform is typically applied to a 2x2 pixel block within an image. Let the pixel values within this block be denoted as $I_{i,j}$, where i and j refer to the row and column positions, respectively. The transform yields four key components: the approximation (or average) coefficient, along with the horizontal, vertical, and diagonal detail components.

Approximation Coefficient

The approximation coefficient A is calculated as the average of all pixel values in the 2x2 block, representing the low-frequency content of the image. The formula for the approximation coefficient is given by:

$$A = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 I_{i,j} \quad (3.1)$$

Horizontal Detail Coefficient

The horizontal detail coefficient H is used to capture the horizontal edges in the image. It is calculated by alternating the sign of the pixel values along each row (i.e., along the columns), based on the column index j :

$$H = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^j I_{i,j} \quad (3.2)$$

Vertical Detail Coefficient

The vertical detail coefficient V highlights the vertical edges in the image. It is computed by alternating the sign of the pixel values along each column (i.e., along the rows), based on the row index i :

$$V = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^i I_{i,j} \quad (3.3)$$

Diagonal Detail Coefficient

The diagonal detail coefficient D is responsible for capturing the diagonal edges in the image. This coefficient is calculated by alternating the sign of the pixel values both row-wise and column-wise, based on the sum of the indices $i + j$:

$$D = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^{i+j} I_{i,j} \quad (3.4)$$

Significance of the $(-1)^{i+j}$ Pattern

The term $(-1)^{i+j}$ plays a key role in the Haar wavelet transform. It alternates the signs of the pixel values, which helps in distinguishing different frequency components. Specifically:

- If $i+j$ is **even**, $(-1)^{i+j} = +1$, meaning the pixel value remains unchanged.
- If $i+j$ is **odd**, $(-1)^{i+j} = -1$, meaning the pixel value is inverted.

This alternating sign pattern enables the Haar transform to effectively capture horizontal, vertical, and diagonal edges in the image.

3.1.3 Graphical Representation of Tetromino Tiling

To illustrate the concept, Figure 3.1 showcases an example of tetromino-based tiling used in tetrolet transformations.

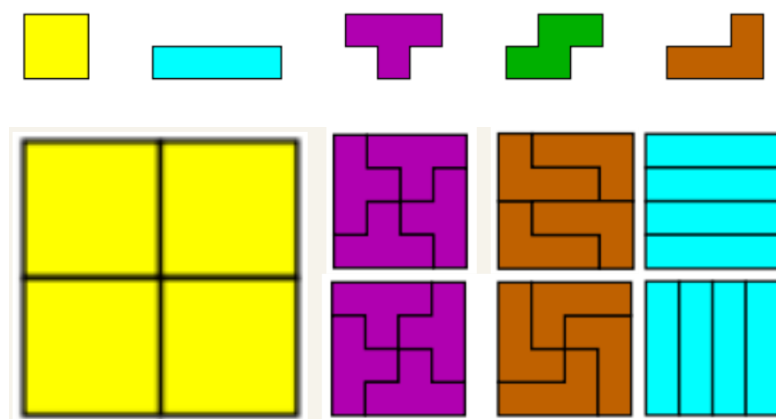


Figure 3.1: Tetromines Tiling Patterns in Tetrolet Transformation

3.1.4 Applications and Future Work

Tetrolet transforms are highly beneficial in image compression, denoising, and pattern recognition tasks. Future research aims to integrate tetrolet-based neural networks for deep learning applications, leveraging their geometric adaptability for feature extraction in convolutional architectures. The fusion of tetrolet transformations with emerging AI techniques could further enhance real-time image processing capabilities.

3.2 Methodology

The methodology for this face image classification problem involves several key steps, starting from image preprocessing to feature extraction and model training. Figure 3.2 depicts the system architecture. The process can be broken down into the following stages:

3.2.1 Dataset Description

The dataset used for this face image classification project is derived from a specialized Indian facial image database that addresses the unique challenges posed by individuals wearing spectacles. Spectacles often cause several issues, such as occlusion, glare, and reflections, which hinder accurate detection and recognition of facial features, especially the eyes. These challenges are collectively termed the "spectacle problem."

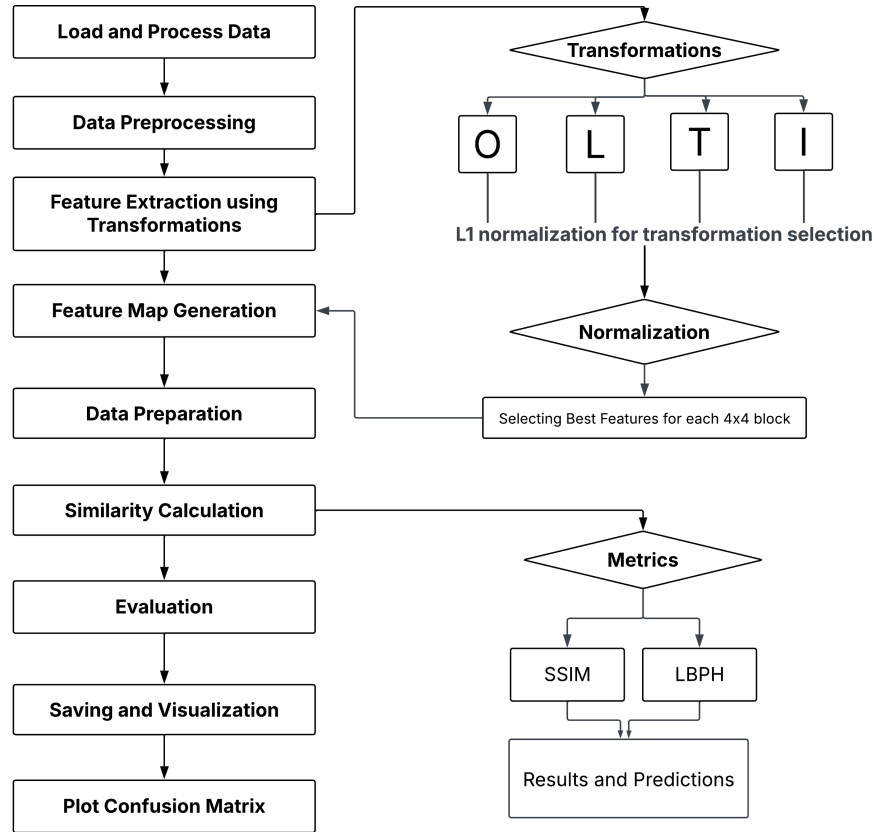


Figure 3.2: System Architecture Overview

Spectacle Problem

The spectacle problem introduces several challenges that interfere with accurate face and eye detection in images. One of the primary issues is occlusion, where the spectacle frames partially or fully obscure facial regions—especially the eyes—making it difficult for detection algorithms to identify crucial features. In addition, glare caused by light reflecting off the lenses can create bright spots in the image, which may hide important facial details such as the eyes and nose. Another complication arises from secondary reflections produced by the lenses themselves, which can distort the visual appearance of the eyes and surrounding areas. These challenges are especially significant in real-world scenarios where individuals frequently wear spectacles, and conventional face detection systems often fail to perform reliably under such conditions.

Indian Facial Image Database

To address these challenges, the Indian facial image database was specifically created to focus on facial images with and without spectacle-related problems.

The database Version 1:

The first release of the dataset contains 271 images that showcase various spectacle-related issues, such as glare and occlusion. These images help train models that can handle the specific problems associated with spectacles.

Metadata and Nomenclature

The dataset is organized using a systematic nomenclature format to categorize each image based on the following parameters:

- NUM: Subject ID Number (from 000 to 270 for Version 1 and a broader range for Version 2).
- G: Gender of the subject (Female = 0, Male = 1).
- S: Spectacles (Without spectacles= 0, With spectacles=1).
- I: Illumination Problem (No illumination problem=0, illumination problem=1).
- O: Occlusion (No occlusion=0, With occlusion = 1).

The database is provided in various formats, including ZIP archives containing image files (in JPG, BMP, and TXT formats), which can be accessed for training and testing purposes. The images are accompanied by their corresponding metadata to assist in understanding and handling the specific challenges posed by spectacles during face recognition tasks.

Data Access and License

The Indian facial image database is publicly available under the Creative Commons Attribution License. The license allows users to freely use and share the data while giving appropriate credit to the creators of the dataset.

Relevance to the Project

This dataset is highly relevant to the current face image classification task, as it provides a realistic set of images with the types of occlusion, glare, and reflections typically encountered in real-world scenarios. The use of this

database allows for better handling of spectacle problems, which is essential for improving face and eye detection accuracy, particularly in cases where spectacles obscure key facial features.

3.2.2 Preprocessing

During the preprocessing phase, several important steps are carried out to prepare face images for further analysis and input into the model. The first step involves converting each image to grayscale. This conversion simplifies the image by removing color information, which reduces computational complexity and allows the model to focus on the structural and textural features of the face. Grayscale images are also more efficient in terms of storage and processing power, which is especially advantageous for real-time or embedded systems.

Next, all images are resized to a consistent 256x256 pixel dimension. This standardization ensures that the model receives input of uniform size, which helps improve the efficiency of processing and enhances the stability during training. These preprocessing steps play a vital role in optimizing the image data for the next stages of face detection and recognition.

3.2.3 Image Block Division

Once the images are preprocessed, they are divided into non-overlapping 4×4 blocks. This allows for local analysis and transformation of each smaller region, enabling the extraction of detailed information from different parts of the image.

Each 4×4 block is further divided into four 2×2 sub-blocks. For each of these 2×2 sub-blocks, the Haar wavelet transform is applied. The Haar wavelet transformation computes the following coefficients:

- A (Approximation): The average of the pixels within the block.
- H (Horizontal Detail): The difference between the pixel values along each row.

- V (Vertical Detail): The difference between the pixel values along each column.
- D (Diagonal Detail): The difference between the diagonal pixel values.

These four coefficients are computed for each 2×2 sub-block and serve as the features for further processing.

3.2.4 Tetrolet Transform

The Tetrolet transform is a block-based technique designed to capture local geometric structures in images more effectively than traditional wavelet methods. It starts by dividing the image into non-overlapping 4×4 pixel blocks. Within each block, the transform rearranges pixels based on optimal tetromino tiling patterns, which draw inspiration from the game Tetris. This approach allows the image to be partitioned adaptively, based on its local structure, enabling the transform to better align with edges and contours within the image.

By dynamically selecting the most suitable tiling configuration that minimizes approximation error, the Tetrolet transform achieves superior energy compaction and sparse representation compared to fixed wavelet bases. This feature is particularly beneficial for preserving local image details such as corners, edges, and textures. The rearranged blocks are then processed further through stages like quantization or compression. Consequently, the Tetrolet transform proves to be an effective pre-processing tool that enhances image representation, making it ideal for tasks in image compression, denoising, and pattern recognition.

3.2.5 Feature Extraction

Feature extraction plays a vital role in image analysis by reducing the image's dimensionality while maintaining essential structural information. In this approach, each 4×4 image block is first divided into four 2×2 sub-blocks. For each sub-block, the Haar wavelet transform is applied to compute four key coefficients: Approximation (A), Horizontal (H), Vertical (V), and Diagonal

(D). These coefficients capture both the broad and fine-grained features of the image, providing a richer representation of texture and edge details.

The approximation coefficient represents the average intensity of the block, while the horizontal, vertical, and diagonal detail coefficients highlight intensity variations in different directions. This multi-directional information is crucial for tasks such as pattern recognition, face detection, and object tracking. Since each 2×2 sub-block generates four coefficients, a single 4×4 block results in 16 feature values. These extracted features create a compact yet detailed feature set, which can be efficiently processed in subsequent stages of machine learning or deep learning models for classification or recognition tasks.

3.2.6 Feature Matrix Construction

The final feature matrix is created by combining the features extracted from all 4×4 blocks throughout the image. Each 4×4 block contributes 16 features, which are derived from the Haar wavelet coefficients of its four 2×2 sub-blocks. By applying this process consistently across the entire image, both global and local variations in texture, intensity, and edge orientation are captured in a well-organized manner. The resulting feature matrix provides a high-dimensional representation that encompasses a comprehensive understanding of the image's content.

This matrix transforms the raw spatial data into a feature space where patterns and relationships between different regions of the image become clearer and more distinguishable. This transformation is particularly useful in machine learning classification tasks, as it allows classifiers to work with meaningful numerical features rather than just raw pixel values. Furthermore, the matrix structure facilitates the use of dimensionality reduction techniques (like PCA or LDA) or normalization methods to improve model performance. Therefore, the feature matrix not only consolidates vital image information but also prepares the data for efficient and effective classification in tasks such as object recognition, face detection, or medical image analysis.

CHAPTER 4

Implementation and Results

The following demonstrates the Tetrolet Wavelet transformation applied to a 256x256 image. The image is resized and converted to grayscale, and then the Haar transformation is applied to each 4x4 block. The three transformations—O-shape, T-shape, I-shape and L-shape—are variations of the Tetrolet transforms applied in different patterns, which results in different ways of dividing and averaging the image data.

4.1 Transformation Process

The Tetrolet wavelet transformation is applied to 4x4 blocks of the image using the following steps: Averaging and differencing across rows and columns for each 4x4 block. Calculating the transformed image with normalized values.

4.1.1 O-Shape Transformation

The O-shape transformation divides the image into 4x4 blocks and applies a Haar-like transformation to each block. Below is the pseudocode for the O-shape transformation, followed by the Python code implementation.

1. For each 4x4 block in the image:
 - (a) Apply the Haar wavelet transform on the block as follows:
 - Average the values of specific cells in the block using Haar coefficients.
 - Normalize the result by dividing by $\sqrt{2}$.
 - (b) Replace the original block with the transformed block.
2. Return the transformed image.

4.1.2 L Transform

The L -transform modifies the image to highlight vertical structures and emphasizes sharp transitions between pixel values. Below is the visual result of applying the L -transform.

4.1.3 T Transform

The T -transform is designed to detect horizontal patterns and smooth transitions across pixels. The transformed image resulting from this process is shown below.

4.1.4 I Transform

The I -transform focuses on extracting and compressing specific details from the image, which enhances its feature representation. The image resulting from the I -transform is displayed below.

4.1.5 L1 Norm Calculation and Transformation Selection

In the code below, the L1 norm is calculated for each transformed block to select the best transformation. The function `calculate_l1_norm()` computes the L1 norm by summing the absolute values of the block, and `select_best_transformation()` selects the best transformation based on the L1 norm.

4.1.6 Computation Using Tetrolet and Haar Transformations

This subsection presents the full implementation of a facial recognition system that compares facial images using both traditional Haar-based detection and advanced Tetrolet transformations. The goal is to compute structural similarity (SSIM) and LBPH (Local Binary Pattern Histogram) similarity across multiple transformed versions of faces for a robust face matching system.

The algorithm includes the following key components:

- Face extraction using Mediapipe and Haar Cascade.
- Feature transformation using four Tetrolet-based encodings: 'O', 'I', 'T', and 'L'.
- Similarity computation using SSIM and LBPH.
- Evaluation using prediction thresholds and confusion matrices.

Below is the complete Python code used in the system:

4.2 Result Analysis

In this section, the results of the block-wise transformation selection strategy applied to a grayscale facial image are presented. The image was divided into non-overlapping 4×4 blocks, and each block was subjected to four different Haar wavelet-based transformations: O, T, L, and I. Each transformation highlights different structural aspects of the image—diagonal, horizontal, vertical, and integrated features—producing distinct outputs for each block.

To determine which transformation best captures the local variations in each block, L1 norm is calculated, which is the sum of absolute pixel differences within the transformed block, for all four transformations. The transformation that resulted in the highest L1 norm was selected as the most effective at capturing the region's details and was used to represent that block in the final feature map.

This adaptive selection process ensures that each part of the image is encoded using the transformation that most accurately reflects its inherent texture, ultimately leading to a more robust representation for tasks such as face recognition.

4.2.1 Block-wise Transformations and Visualization

In this section, the visual outcomes are presented and analyzed, of applying four distinct transformations—O, L, T, and I—to each 4×4 block of the input grayscale facial image. These transformations, based on variations of the Haar

wavelet technique, emphasize different structural features such as diagonals, verticals, horizontals, or combinations thereof.

The application of these transformations alters the pixel arrangements within each block, effectively highlighting specific directional patterns and localized textures. The resulting images not only showcase the impact of each transformation on facial structures but also provide visual cues to help understand the enhanced feature representations. For each transformation, a corresponding 16×16 block-level visualization is included, showing how local texture variations propagate across the entire image grid.

These visual results demonstrate the ability of wavelet-based transformations to capture subtle facial features that are crucial for accurate recognition, underscoring their value in feature extraction for tasks like face detection.

Transformation O Results

Transformation O applies a basic Haar wavelet transformation to non-overlapping 4×4 blocks of the grayscale facial image. The purpose of this transformation is to decompose local pixel intensities into frequency sub-bands, capturing both coarse and fine structural information within each block.

Notably, the Haar transform in this context emphasizes contrast variations along the diagonals. This results in visible alterations that are most pronounced in the diagonal regions of the transformed image. The diagonal detail coefficients (D) extracted during the Haar process are particularly sensitive to diagonal edge patterns and intensity transitions. As a result, the block-wise transformation highlights subtle changes in facial contours, textures, and edge orientations, which are crucial for distinguishing between different faces.

The Figure 4.1 illustrates the visual effect of Transformation O, where the diagonal emphasis becomes more apparent. These diagonal details provide valuable features for subsequent similarity comparisons and classification tasks.

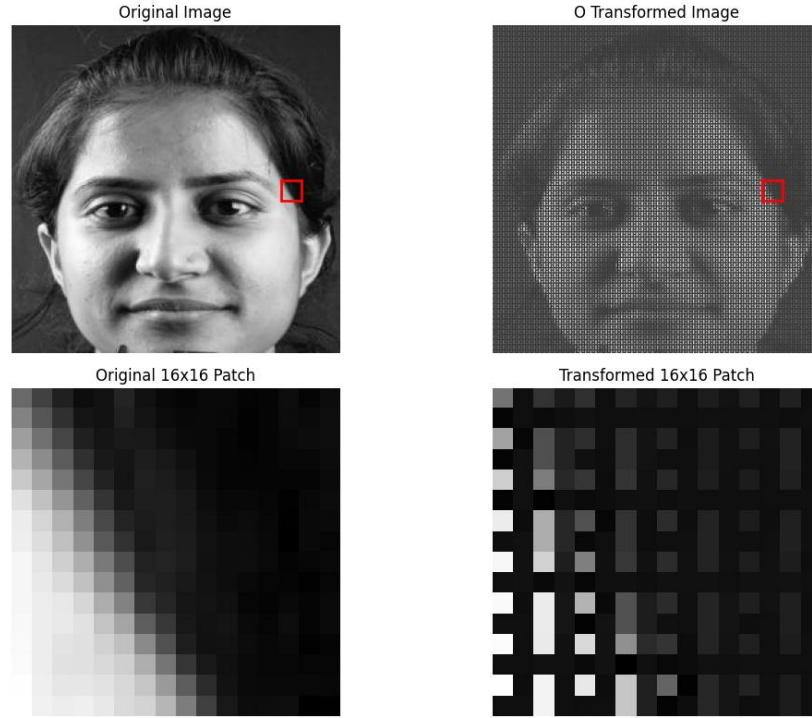


Figure 4.1: O Transformation Results

Transformation L Results

Transformation L applies a modified wavelet transform to each 4×4 block, focusing on extracting vertical detail coefficients from the Haar decomposition. Unlike Transformation O, which accentuates diagonal features, Transformation L targets vertical intensity gradients by analyzing changes along the columns of each block. This transformation effectively highlights vertical edges and linear structures, making it particularly useful for emphasizing facial features like the nose bridge, lips, or the outline of the face, which often display distinct vertical characteristics. The resulting transformed image reflects these enhancements, creating contrast in vertical regions and aiding in better feature differentiation during the classification process. The Figure 4.2 shows the visual effect of Transformation L, where the vertical details are amplified, providing a clearer structural representation of the face.

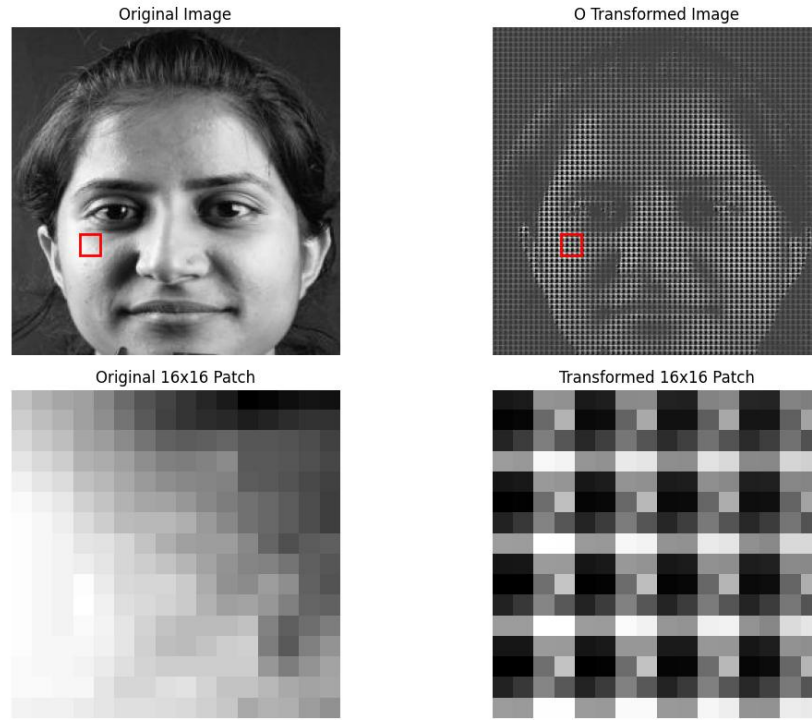


Figure 4.2: L Transformation Results

Transformation T Results

Transformation T is designed to combine both horizontal and vertical detail components from the Haar wavelet transform, applied to each 4×4 block of the grayscale facial image. This dual approach allows the model to capture a broader set of features by emphasizing transitions along both axes. As a result, the transformation highlights structural patterns like facial outlines, the contours of the eyes and brows, and lip curvature — features that exhibit both horizontal and vertical variations. By integrating these directional coefficients, Transformation T introduces more complexity into the block-wise representation, producing a texture-rich output that is especially useful for distinguishing subtle facial differences. The Figure 4.3 shows how this transformation creates intricate patterns, reflecting multi-directional facial information that is crucial for accurate face matching.

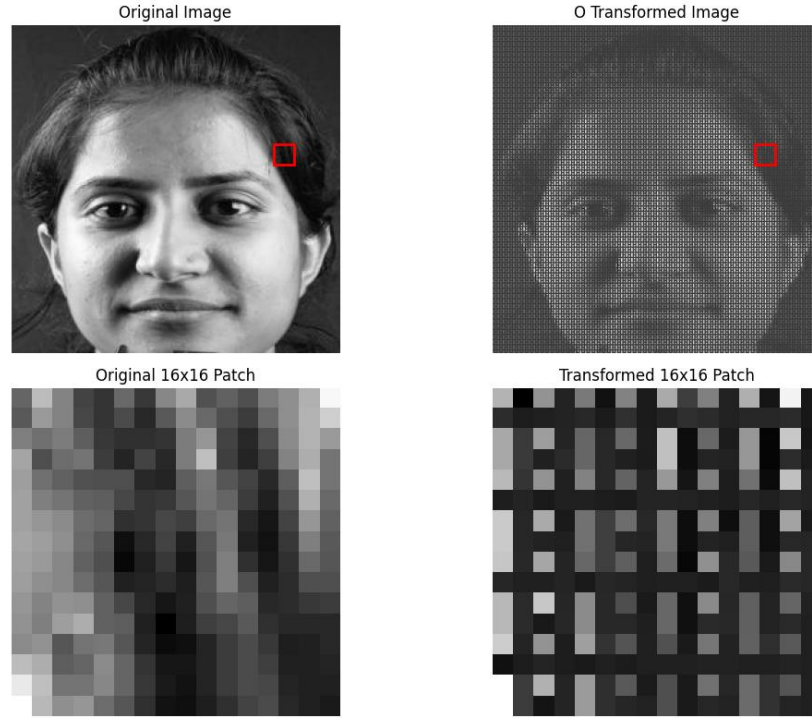


Figure 4.3: T Transformation Results

Transformation I Results

Transformation **I** integrates both horizontal and vertical Haar wavelet components in a comprehensive manner, resulting in the most detailed and complex transformation among all the variants. Applied on each 4×4 block of the grayscale facial image, this transformation extracts not only the basic approximation coefficients but also blends the vertical (V) and horizontal (H) details to form a densely encoded representation. The fusion of these directional features captures fine-grained facial textures, symmetrical features, and edge orientations that are critical for distinguishing between individuals, especially in scenarios involving subtle variations. The increased intricacy of the block-wise transformation allows for better retention of localized patterns while enhancing sensitivity to both axis-aligned and oblique structures. As a result, Transformation I produces a highly informative feature map that significantly boosts the discriminative power of the overall system. The Figure 4.4 visually reflects the complexity introduced by this transformation across facial regions. This transformation is shown below.

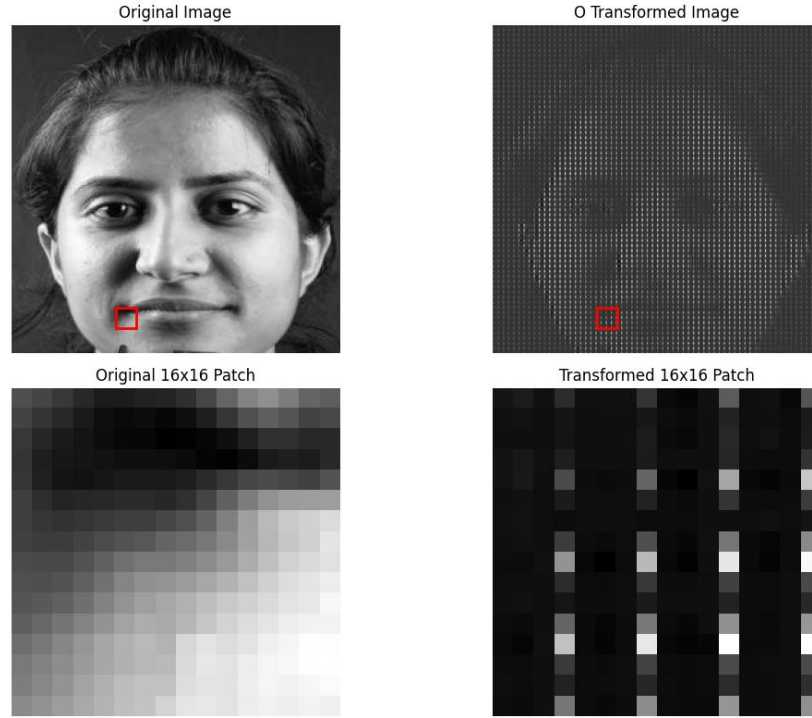


Figure 4.4: I Transformation Results

These transformations demonstrate how different Haar wavelet techniques can influence the visual structure of the image at the block level, highlighting different coefficients and areas of change.

4.2.2 Transformation for Each Block

The transformation selected for each 4x4 block is shown below. The matrix represents the transformation applied to each block in the image:

$$\begin{bmatrix} L & L & L & L & \dots & L & L & L \\ L & L & L & L & \dots & L & L & L \\ L & L & L & L & \dots & L & L & L \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ L & L & L & L & \dots & O & L & L \\ L & L & L & L & \dots & I & L & L \end{bmatrix}$$

Each entry in the transformation matrix corresponds to the best transformation selected for the respective 4x4 block.

Sample Image After Transformation

Figure 4.5 is the sample image after applying the best transformations for each block. The transformations enhance the image in various ways based on their L1 norm calculations.

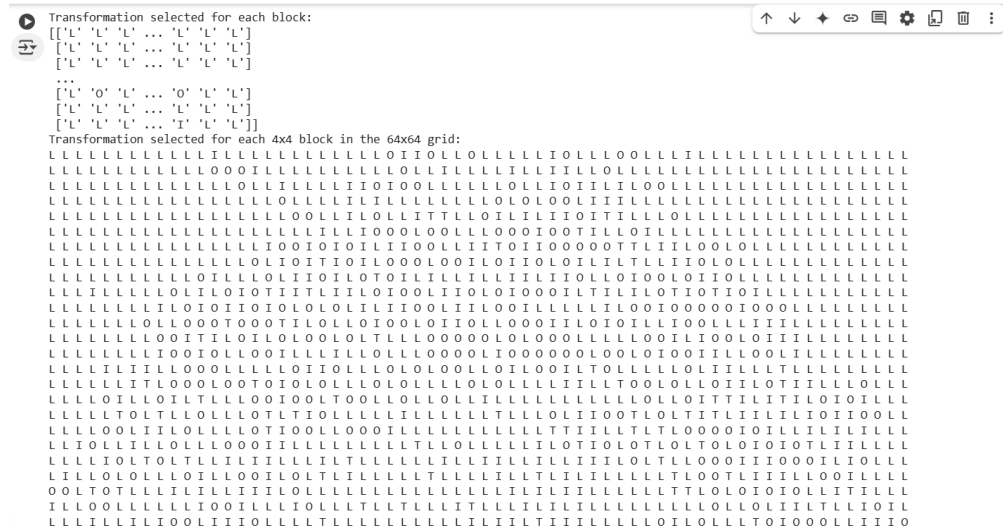


Figure 4.5: Applying the best transformation to each block.

The block-wise analysis demonstrated that the transformation L was the most frequently selected for the given image, with several regions benefiting from other transformations like O, I, and T. This suggests that L is likely the most suitable transformation for the specific features in the image. The visual results confirm these findings, with the transformed image exhibiting improvements in clarity and contrast where appropriate.

4.2.3 Prediction Results

This work presents a face recognition pipeline based on facial similarity analysis using MediaPipe, grayscale transformations, and Structural Similarity Index Measure (SSIM). The objective is to predict whether two facial images belong to the same person.

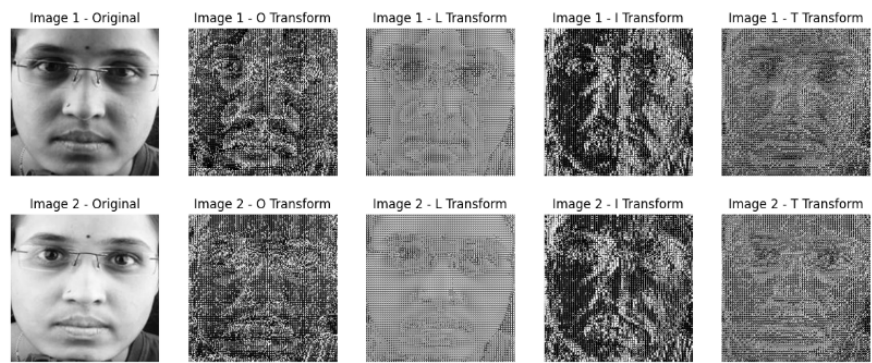
Face Extraction: The process begins with identifying and extracting facial regions from the input images using the FaceDetection module provided by the MediaPipe library. The images are first converted from their original BGR

format to the RGB color space, as required for MediaPipe’s internal processing. After the conversion, the face detection module locates facial landmarks and computes a bounding box around the face. This bounding box is used to crop the face, separating it from the background and any irrelevant elements. The cropped face is then converted into grayscale by combining the RGB channel values in a way that preserves the luminance necessary for structural analysis. Lastly, the grayscale image is resized to a uniform resolution of 256×256 pixels to ensure consistency across the dataset, facilitating efficient feature extraction and comparison in later stages of the recognition system.

Image Transformation: Once the grayscale facial image is resized to a standardized resolution, it undergoes a series of four custom transformations: O, L, I, and T. Each transformation is specifically designed to capture different structural or textural elements of the facial region. Transformation O highlights diagonal features using basic Haar decomposition, while L focuses on extracting vertical details. T emphasizes both vertical and horizontal gradients, resulting in a more complex pattern representation, and I combines all of these components to create a highly detailed image. These transformations work together to bring out subtle differences between faces by highlighting various directional textures. The outcome is a set of images, including the original grayscale version and its four transformed counterparts, which together form a rich multi-view feature map. This diverse set of features enhances the accuracy and reliability of the similarity comparison process in face recognition.

Similarity Measurement: To compare the corresponding transformations of the two input faces, SSIM is computed for the following image pairs. Prediction results are shown in the Figure 4.6 and 4.7:

- O Transform vs O Transform
- L Transform vs L Transform
- I Transform vs I Transform
- T Transform vs T Transform



Similarity Scores:

Original vs Original: 0.5737

O Transform vs O Transform: 0.2653

L Transform vs L Transform: 0.5653

I Transform vs I Transform: 0.2821

T Transform vs T Transform: 0.4392

Threshold Check:

Original vs Original: Score = 0.5737, Threshold = 0.40, Passed = True

O Transform vs O Transform: Score = 0.2653, Threshold = 0.10, Passed = True

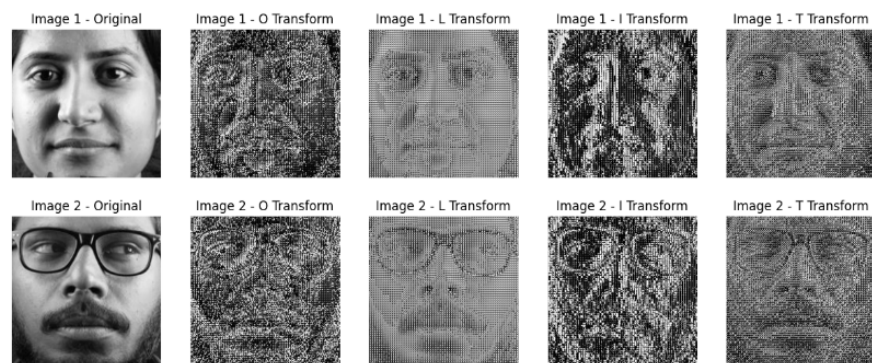
L Transform vs L Transform: Score = 0.5653, Threshold = 0.50, Passed = True

I Transform vs I Transform: Score = 0.2821, Threshold = 0.10, Passed = True

T Transform vs T Transform: Score = 0.4392, Threshold = 0.30, Passed = True

✓ Face Recognized: Images likely belong to the same person.

Figure 4.6: Prediction - Face Match



Similarity Scores:

Original vs Original: 0.2205

O Transform vs O Transform: 0.0736

L Transform vs L Transform: 0.4948

I Transform vs I Transform: 0.0775

T Transform vs T Transform: 0.2679

Threshold Check:

Original vs Original: Score = 0.2205, Threshold = 0.40, Passed = False

O Transform vs O Transform: Score = 0.0736, Threshold = 0.10, Passed = False

L Transform vs L Transform: Score = 0.4948, Threshold = 0.50, Passed = False

I Transform vs I Transform: Score = 0.0775, Threshold = 0.10, Passed = False

T Transform vs T Transform: Score = 0.2679, Threshold = 0.30, Passed = False

✗ Face does NOT Match: Images likely belong to different people.

Figure 4.7: Prediction - Face does not match

SSIM is a perceptual metric that measures the similarity between two images based on luminance, contrast, and structure. A score closer to 1 implies higher similarity.

Threshold-Based Classification: Each transformation pair, derived from the original and altered facial images, is assessed using the Structural Similarity Index Measure (SSIM), which evaluates the visual similarity between two images based on luminance, contrast, and structure. For every pair, a similarity threshold is predefined based on prior testing or validation using sample data. When the SSIM value for a particular transformation exceeds its threshold, the comparison is considered a match, suggesting a strong resemblance between the facial features in that transformation. This matching process is repeated for all five transformation pairs. To enhance accuracy and reliability in the decision-making process, the system uses a majority-vote approach: if at least four out of the five transformation pairs meet the similarity threshold, the system concludes that the two input images belong to the same individual. On the other hand, if fewer than four pairs satisfy the similarity criteria, the system classifies the faces as belonging to different individuals, indicating distinct structural or textural patterns.

Visualization: The prediction results are visualized using side-by-side plots of the transformed faces. Each row displays the original and transformed outputs for the respective image. A sample result is shown in Figure 4.7.

4.3 Evaluation Results

This section outlines the methodology used to evaluate facial similarity between image pairs, incorporating both traditional face recognition techniques and transformation-based methods. The evaluation begins with a dataset of image pairs, each associated with a ground truth label that determines whether the pair represents a match or not. The images were processed using two different face detection techniques:

- The first technique uses Haar Cascade face detection, where faces are detected using OpenCV's Haar Cascade Classifier. The cropped faces are resized to a fixed resolution of 256×256 pixels.
- The second technique employs MediaPipe-based face detection, where the faces are extracted and converted into grayscale images. These images undergo four directional transformations: O, I, T, and L, which capture various features of the face from different angles.

4.3.1 Similarity Metrics

To quantify the similarity between the face pairs, two primary metrics are utilized:

1. **Structural Similarity Index (SSIM)**: SSIM measures the perceptual similarity between two images by comparing their luminance, contrast, and structure. The SSIM formula is given by:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4.1)$$

Where: - μ_x, μ_y are the mean pixel intensities of images x and y , - σ_x^2, σ_y^2 are the variances of x and y , - σ_{xy} is the covariance between the two images, - C_1 and C_2 are small constants to stabilize the calculation when the denominator is small.

2. **Local Binary Pattern Histogram (LBPH)**: LBPH computes a binary pattern for each pixel in a face image and then generates a histogram of these patterns. The similarity is computed by comparing the histograms of two images. The LBPH formula is given by:

$$LBPH(x, y) = \sum_{i=0}^{n-1} |p_i - q_i| \quad (4.2)$$

Where: - p_i and q_i are the pixel values at the i -th position in the training and testing images, respectively, - n is the total number of pixels in the local region used for comparison.

4.3.2 Prediction Strategy

For the MediaPipe-based method, similarity scores are computed for each transformation (O, I, T, L) using SSIM and LBPH metrics. A decision-making mechanism is applied to determine the final prediction:

- A predefined threshold is applied to each transformation's SSIM and LBPH score.
- If at least three of the four transformations meet the threshold, the image pair is classified as a match.

For Haar-based face detection, the evaluation thresholds are as follows:

- SSIM threshold: 0.4
- LBPH threshold: 0.6

Predictions are recorded and stored in a CSV file for further analysis.

The performance of each model is evaluated by comparing the predicted labels with the actual ground truth labels. Key evaluation metrics such as accuracy, precision, recall, and F1-score are calculated, alongside confusion matrices to assess the overall performance. The accuracy formula is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.3)$$

Where: - TP = True Positives, TN = True Negatives, - FP = False Positives, FN = False Negatives.

The results of the comparison reveal that the MediaPipe-based method, incorporating high-resolution image transformation and multi-directional comparison, outperforms the Haar-based method, which serves as a baseline.

4.3.3 Confusion Matrices and Results

The confusion matrices and evaluation metrics for both Haar and Tetralet transformation methods are summarized in Table 4.1. The performance comparison is illustrated using SSIM and LBPH methods.

Table 4.1: Accuracy Comparison for Tetrolet and Haar Methods for SSIM and LBPH.

Method	SSIM Accuracy (%)	LBPH Accuracy (%)
Tetrolet Transformation	95.54	81.87
Haar Classifier	90.82	89.99

Figures 4.8 and 4.9 present the confusion matrices for the Tetrolet method with SSIM and LBPH, respectively. Similarly, Figures 4.10 and 4.11 show the results for the Haar method.

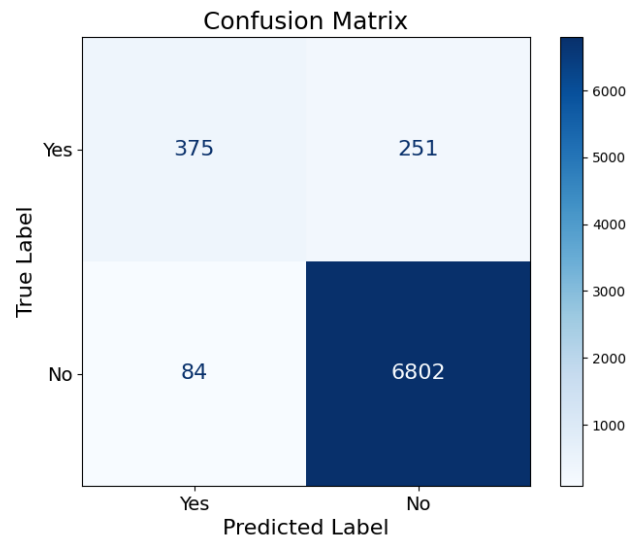


Figure 4.8: Tetrolet SSIM Confusion Matrix

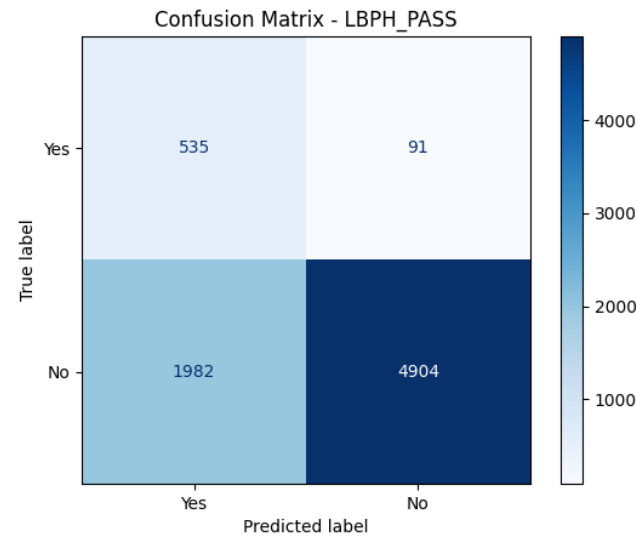


Figure 4.9: Tetrolet LBPH Confusion Matrix

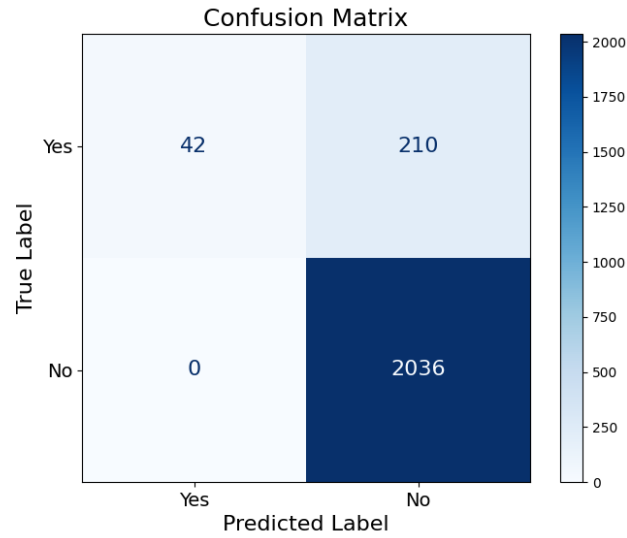


Figure 4.10: Haar SSIM Confusion Matrix

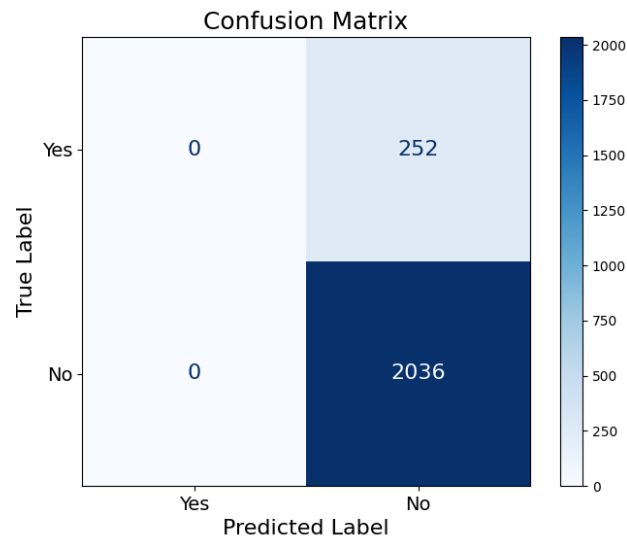


Figure 4.11: Haar LBPH Confusion Matrix

4.4 Database Schema

MongoDB Compass is a graphical user interface for MongoDB that allows developers to explore and interact with their MongoDB data. While not a testing tool per se, it can be used to visually inspect and validate data in the MongoDB database.

The MongoDB collection depicted as in Table 4.2 represents student profiles, where each document holds comprehensive details. The automatically

Table 4.2: Student Registrations Data

studentName	rollNo	branch	section	semester	v	year
B Deekshitha	1.60E+11	it	H1	v	0	3
Poojitha Tanguturi	1.60E+11	it	H1	v	0	3
Gadda Bhavya Shree	1.60E+11	it	H1	v	0	3
Dama Pranati Sreya	1.60E+11	it	H1	v	0	3
Dama Vinayak Snehit	1.60E+11	bio-tech	K	iii	0	2

generated Object Id "id" functions as the primary key, ensuring document uniqueness. Student information includes the "student Name," "roll No," "branch," "section," "semester," "phone Number," "personal Email," "official Email," "parent Name," "parent Phone Number," and "parent Email," all stored as Strings. Additional fields encompass the "password," "confirm Password," and "registration Date," representing security credentials and the date of registration. The "v" field denotes the version and is set to 0. The "year" field signifies the academic year and is stored as a String. This structured MongoDB collection efficiently manages diverse student data, facilitating organized retrieval and modification of individual profiles.

Table 4.3: Student Gatepass Details in collection

rollNo	reason	date	timeSlot	status
1.60E+11	health issue	14-11-2023	12:15-01:00	Approved
1.60E+11	Free Periods	21-11-2023	01:00-02:00	Approved
1.60E+11	No classwork	07-11-2023	12:15-01:00	Approved
1.60E+11	free Hours	21-11-2023	02:00-03:00	Pending
1.60E+11	sick	21-11-2023	10:10-11:10	Approved
1.60E+11	free periods	22-11-2023	02:00-03:00	Approved
1.60E+11	free periods	22-11-2023	12:15-01:00	Rejected
1.60E+11	medical purpose	25-11-2023	11:10-12:15	Pending
1.60E+11	free periods	24-11-2023	02:00-03:00	Rejected
1.60E+11	free periods	25-11-2023	11:10-12:15	Pending
1.60E+11	free periods	28-11-2023	02:00-03:00	Approved
1.60E+11	free hours	29-11-2023	02:00-03:00	Approved
1.60E+11	free periods	28-11-2023	02:00-03:00	Pending
1.60E+11	sick	29-11-2023	02:00-03:00	Pending
1.60E+11	free periods	13-12-2023	12:15-01:00	Rejected
1.60E+11	no classes	15-12-2023	01:00-02:00	Pending

The MongoDB document shown in Table 4.3 represents a record of a student's leave request. The unique identifier, "id," is an automatically generated Object Id. The "roll No" field contains the student's roll number, "reason" provides details on the reason for the leave (in this case, a "health issue"). The "date" field specifies the requested leave date, and "time Slot" indicates the preferred time period for the leave, set here as "12:15-01:00". The "status" field is currently marked as "Pending," representing the current state of the leave request. The "v" field denotes the version and is set to 0. This MongoDB document organizes information related to student leave requests, facilitating efficient tracking and processing.

Table 4.4: Security details in collection

Password	Email	PhoneNumber	UserName
123456	cbitsecurity@gmail.com	9999999999	CBITSecurity

The MongoDB collection represented in Table 4.4 is designed to store user profiles specifically for the college security personnel. Each document in the collection is uniquely identified by an automatically generated ObjectId, referred to as "id." The document contains essential user information, including "Password," "Email," "Phone Number," and "User Name." The "Password" field securely stores the user's login credentials, while the "Email" field holds the individual's email address. The "Phone Number" field captures the contact number, and the "User Name" is the unique identifier assigned to each security personnel member. This well-structured collection allows for secure user authentication and efficient management of personal and contact details, with the ObjectId ensuring each document's uniqueness.

Table 4.5: Faculty Details in collection

Name	EmployeeID	Department	Year	Class	Section
Gadda Bhavya Shree	1601008	Civil	2	bio-tech	K
Bandam Deekshitha	1601002	CSE	4	Civil	A1
Dama Pranati Sreya	1601006	IT	3	IT	H1
Tanguturi Poojitha	1601021	IT	1	IT	H1

The MongoDB collection as shown in Table 4.5 stores detailed individual profiles, with an automatically generated ObjectId ("id") serving as the primary key. It includes fields such as "Name," "Employee ID," and "Department,"

all represented as Strings, while "Year" is stored as an Integer. Academic information, including "Class" and "Section," is also recorded as Strings. Contact details, such as "Personal Mail," "Official Mail," and "Phone Number," are stored as Strings. The "password" field ensures secure access to accounts. This well-organized structure supports efficient storage and retrieval of diverse personal and academic data, making the collection a versatile tool for managing individual profiles.

4.5 User Interface Screens

4.5.1 Home Interface

Figure 4.12 is The SMART EXIT Home page introducing an intuitive online gatepass system, providing distinct login options for students, faculty, and security. Streamlined with a user-friendly design, it efficiently guides users to their respective interfaces, emphasizing functionality for seamless departure management within the academic setting.

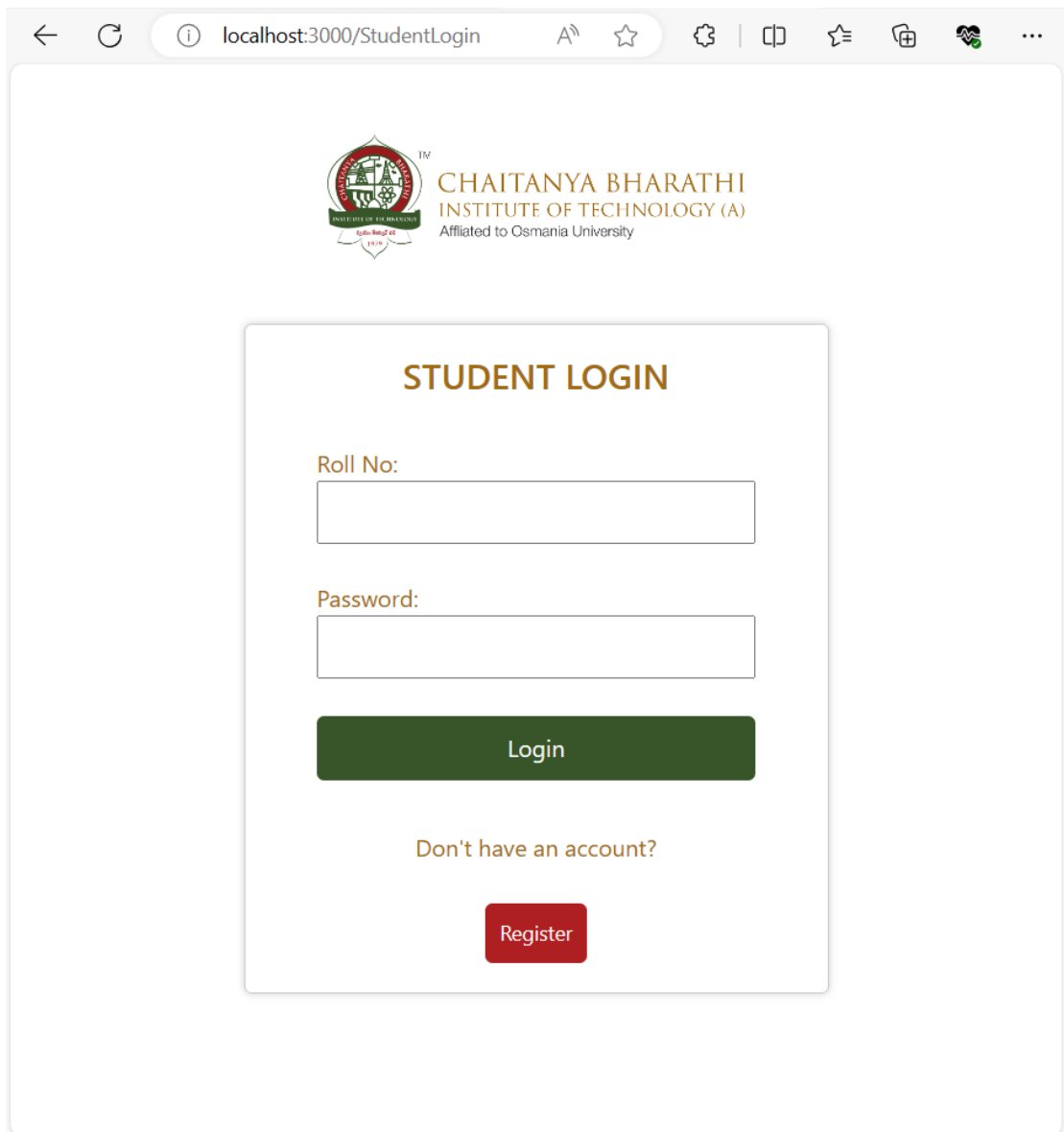


Figure 4.12: Home Page

This user-friendly prompt outlines the necessary steps for users to navigate the interface, guiding them through the login and registration process to gain access to the system.

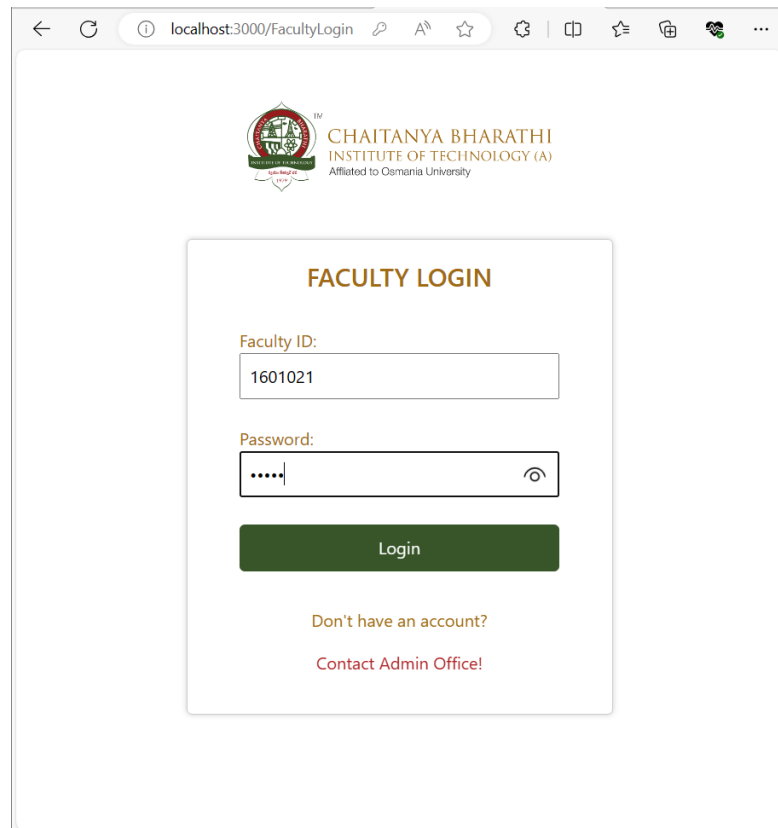
4.5.2 Login Interfaces

In the login pages, Figures 4.13, 4.14, and 4.15, upon successful entry of valid credentials and clicking the "Login" button, users are seamlessly directed to the respective Student, Faculty, and Security main pages. In contrast, incorrect login credentials prompt an alert message such as "Security Not Found." Additionally, for Faculty and Security users attempting to log in without valid credentials, a prompt encourages them to contact the admin office for account creation, and student login has a redirection to student registration if the student is not registered.



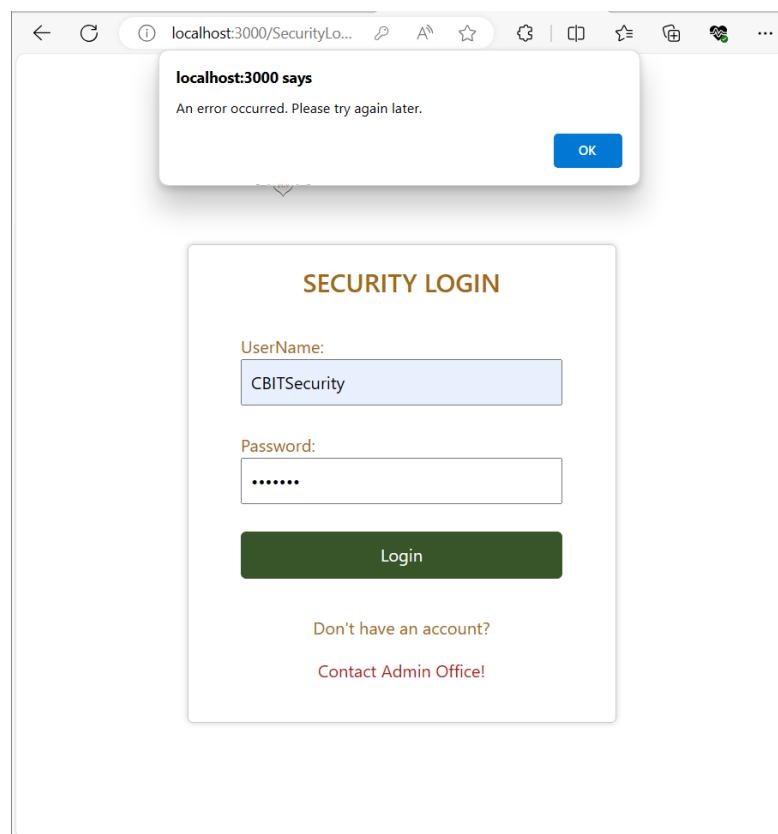
The screenshot shows a web browser window with the address bar displaying "localhost:3000/StudentLogin". The page features the logo of Chaitanya Bharathi Institute of Technology (A) Affiliated to Osmania University. The main content area is titled "STUDENT LOGIN" and contains two input fields: "Roll No:" and "Password:". Below these fields is a green "Login" button. At the bottom, there is a link "Don't have an account?" and a red "Register" button.

Figure 4.13: Student Login Page



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/FacultyLogin'. The page features the logo of Chaitanya Bharathi Institute of Technology (A) at the top left, which is affiliated to Osmania University. The main heading is 'FACULTY LOGIN'. Below this, there are two input fields: 'Faculty ID:' with the value '1601021' and 'Password:' with masked characters '....'. A green 'Login' button is positioned below the password field. At the bottom of the login form, there are two links: 'Don't have an account?' and 'Contact Admin Office!'.

Figure 4.14: Faculty Login Page



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/SecurityLo...'. An error message box is visible at the top, stating 'localhost:3000 says' and 'An error occurred. Please try again later.' with an 'OK' button. The main heading is 'SECURITY LOGIN'. Below this, there are two input fields: 'UserName:' with the value 'CBITSecurity' and 'Password:' with masked characters '.....'. A green 'Login' button is positioned below the password field. At the bottom of the login form, there are two links: 'Don't have an account?' and 'Contact Admin Office!'.

Figure 4.15: Security Login Page

4.5.3 Student Registration Interface

The Student Registration page Figure 4.16 facilitates user-friendly registering of new students in the SMART EXIT system. The form includes essential fields such as name, roll number, branch, year, section, semester, contact details, parent info, and password. Users select options from dropdowns for accurate data entry. The clean layout and subtle hover effects on the "Submit" button ensure an intuitive and secure registration process, aligning with SMART EXIT's overall design philosophy.

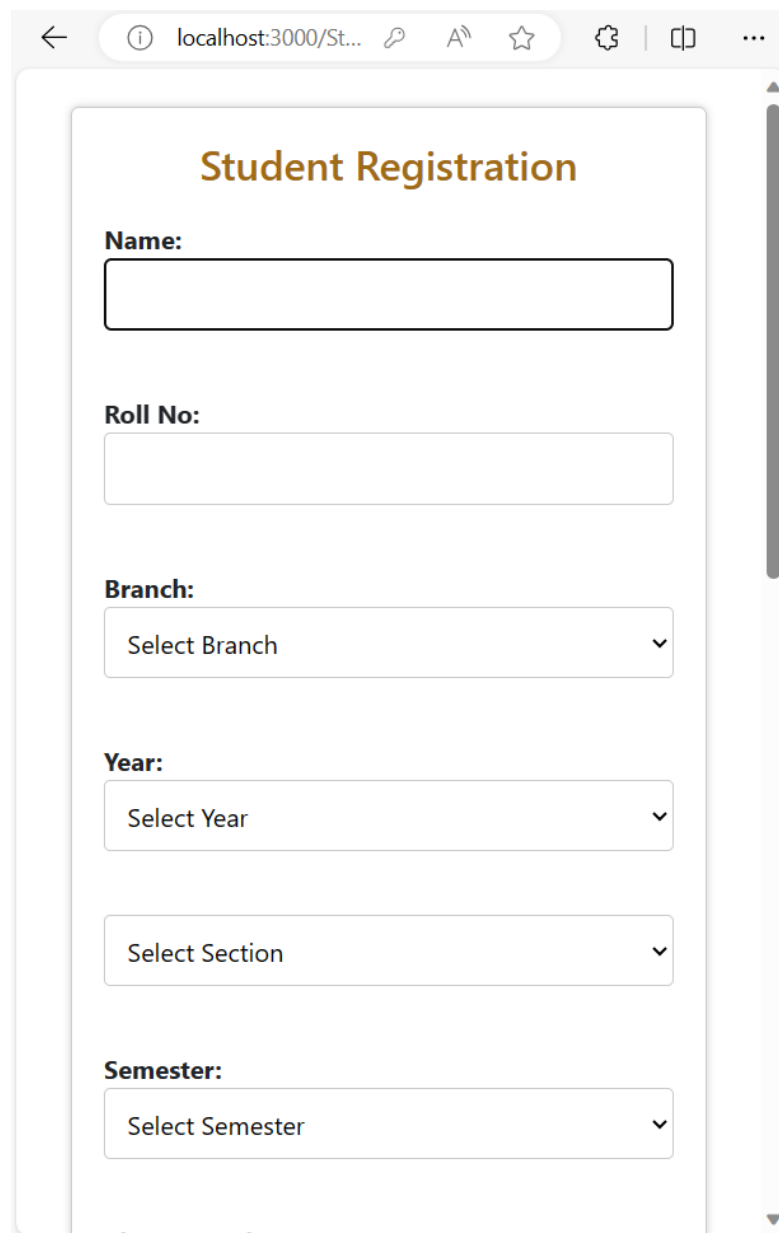
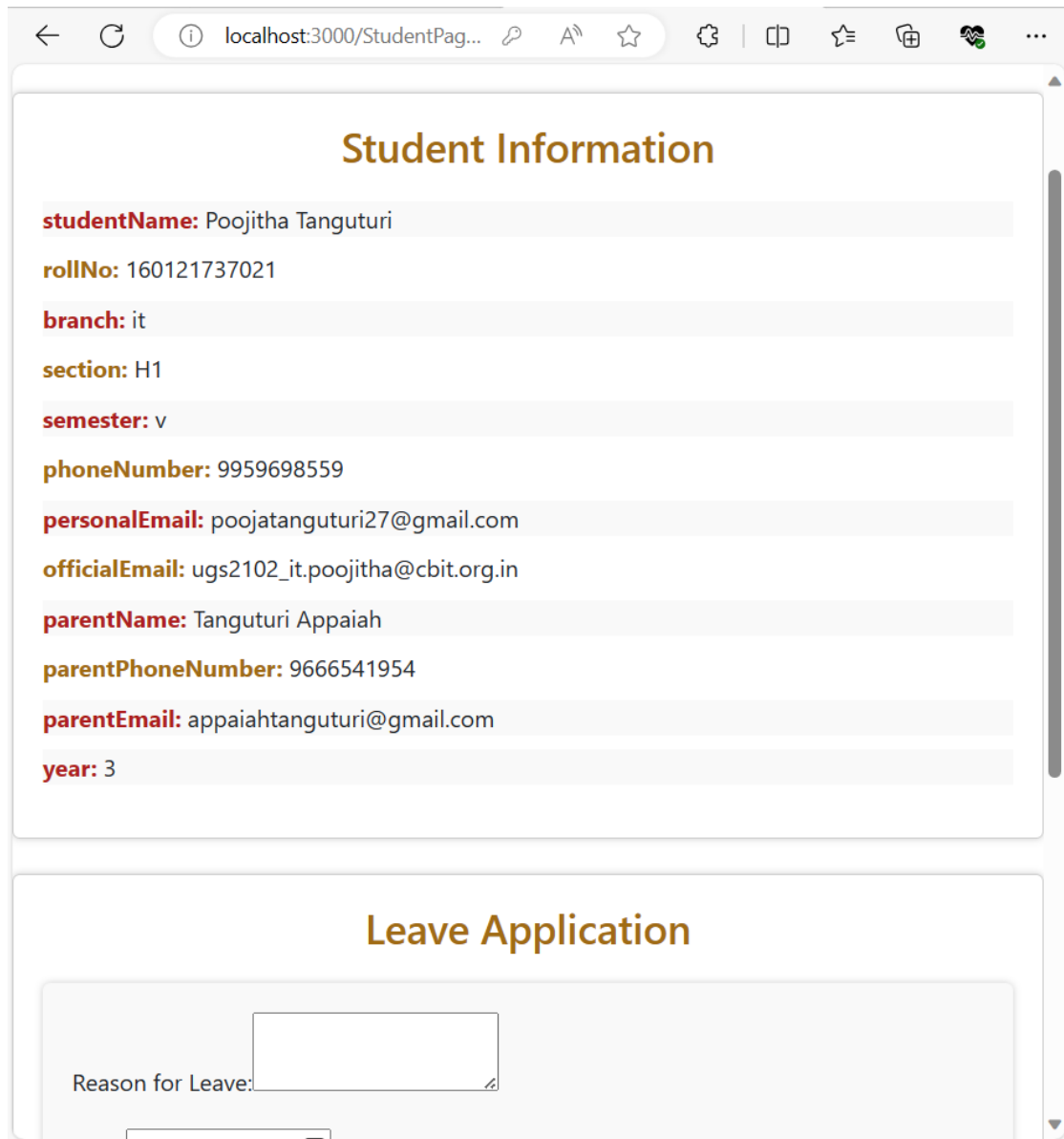
A screenshot of a web browser displaying the "Student Registration" page. The browser's address bar shows "localhost:3000/St...". The page has a light gray background with a white registration form in the center. The form is titled "Student Registration" in a bold, orange font. It contains several input fields and dropdown menus: a text field for "Name:", a text field for "Roll No:", a dropdown menu for "Branch:" with the placeholder "Select Branch", a dropdown menu for "Year:" with the placeholder "Select Year", a dropdown menu for "Section:" with the placeholder "Select Section", and a dropdown menu for "Semester:" with the placeholder "Select Semester". Each dropdown menu has a small downward arrow icon on the right. The form is surrounded by a subtle shadow, and there is a vertical scrollbar on the right side of the browser window.

Figure 4.16: Student Registration Page

4.5.4 Student Dashboard

The Student Page presents a comprehensive view of the student's information and facilitates leave application management. The top section Figure 4.17 displays essential student details such as name, roll number, branch, year, section, semester, contact information, and parent details in an organized list. The alternating background colors enhance readability.



The screenshot shows a web browser window with the address bar displaying 'localhost:3000/StudentPag...'. The page is divided into two main sections. The top section, titled 'Student Information' in a bold, brown font, contains a list of student details with alternating light gray and white background colors for each row. The details include: studentName: Poojitha Tanguturi, rollNo: 160121737021, branch: it, section: H1, semester: v, phoneNumber: 9959698559, personalEmail: poojatanguturi27@gmail.com, officialEmail: ugs2102_it.poojitha@cbit.org.in, parentName: Tanguturi Appaiah, parentPhoneNumber: 9666541954, parentEmail: appaiahtanguturi@gmail.com, and year: 3. The bottom section, titled 'Leave Application' in a bold, brown font, contains a form with a text input field labeled 'Reason for Leave:' and a small icon of a document with a pencil.

Field	Value
studentName	Poojitha Tanguturi
rollNo	160121737021
branch	it
section	H1
semester	v
phoneNumber	9959698559
personalEmail	poojatanguturi27@gmail.com
officialEmail	ugs2102_it.poojitha@cbit.org.in
parentName	Tanguturi Appaiah
parentPhoneNumber	9666541954
parentEmail	appaiahtanguturi@gmail.com
year	3

Leave Application

Reason for Leave:

Figure 4.17: Student Page

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localhost:3000/StudentPag...

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year: 5

Leave Application

Reason for Leave:

Date:

dd-mm-yyyy

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Time Slot:

12:15-01:00

▼

Submit Leave Application

Previous Leave Applications

Index	Reason	Date	Time Slot	Status
1	sick	2023-11-21	10:10-11:10	Approved
2	free periods	2023-11-22	02:00-03:00	Approved
3	free periods	2023-12-13	12:15-01:00	Pending
4	no classes	2023-12-15	01:00-02:00	Pending

Figure 4.18: Student Leave Applications

Furthermore, the page offers a historical overview of previous leave applications. A table summarizes each application, detailing the reason, date, time slot, and its current status as shown in Figure 4.18. This provides students with a transparent record of their leave history, aiding in tracking and management. In the absence of previous applications, a concise message indicates the absence of recorded leave instances.

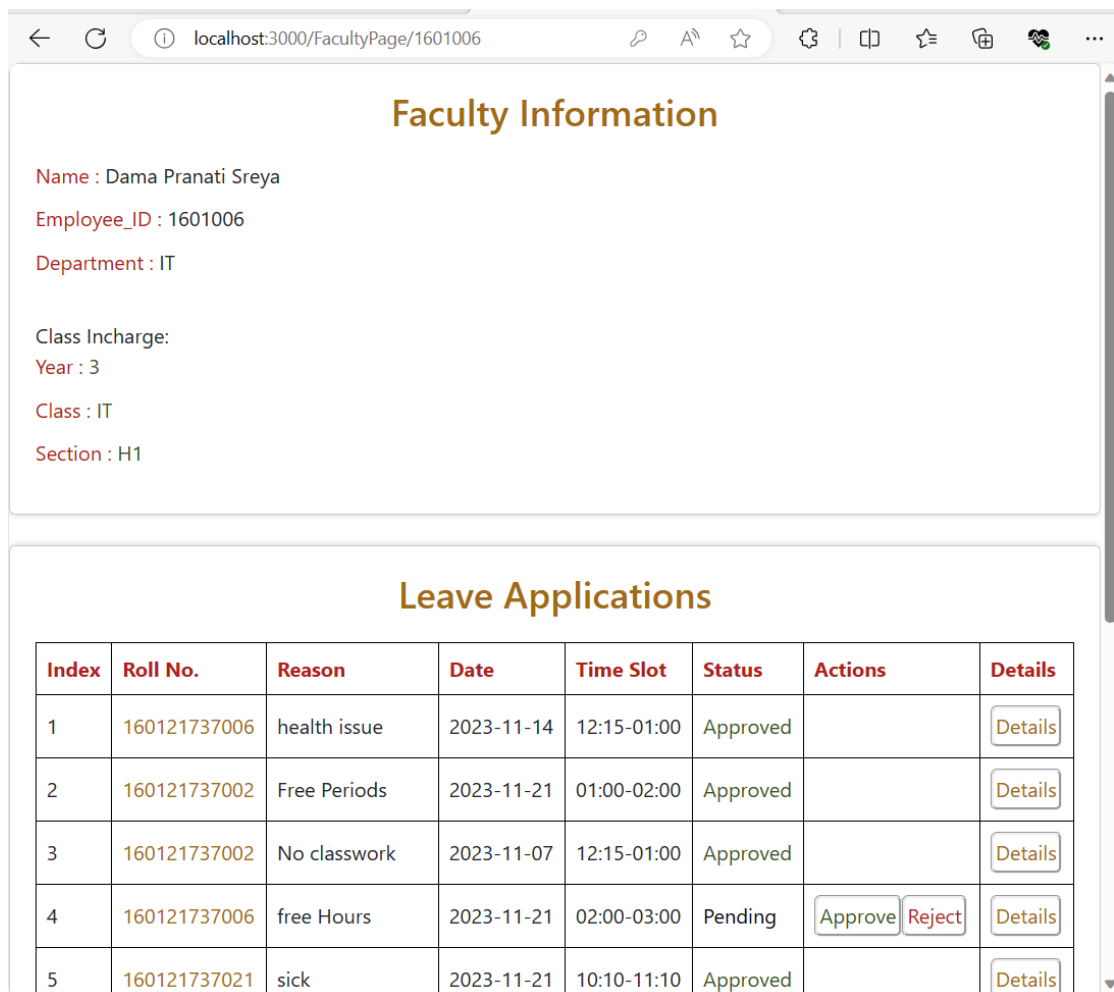


Figure 4.19: Faculty Dashboard

4.5.5 Faculty Dashboard

The Faculty Page Figure 4.19 provides faculty members with a user-friendly interface to access essential information and manage leave applications efficiently. It displays important details such as the faculty member's name, class in charge status, academic year, class, and section. The information is presented in a well-organized list format, with color-coded elements used to improve visual clarity and enhance readability.

Beneath the faculty information section, Figure 4.21, an easy-to-use table shows the leave applications submitted by students. This table includes key details such as the student's roll number, the reason for the leave, the date, the time slot, the application status, and actionable buttons for approving or rejecting pending applications. The color-coded status system helps quickly identify the current state of each application. Faculty members also have the

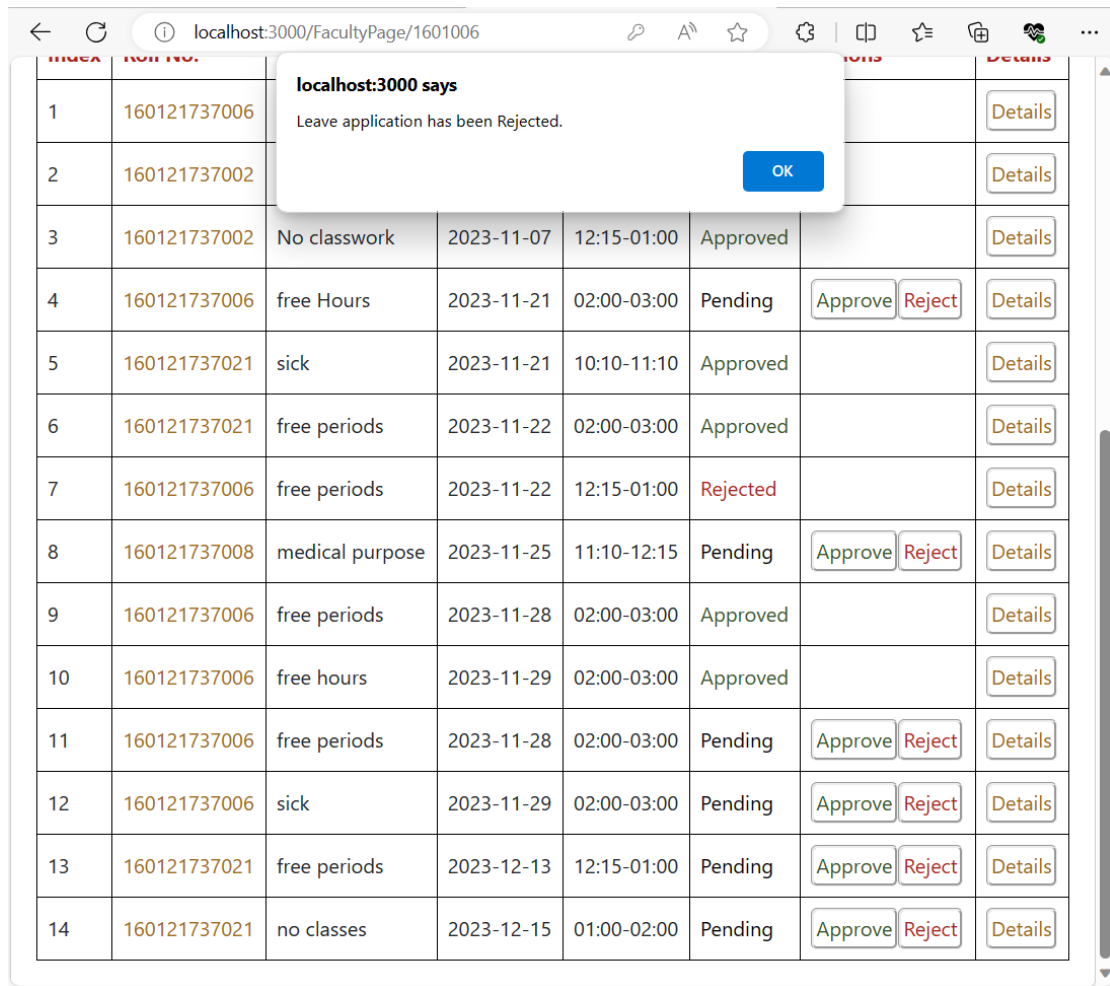


Figure 4.20: Faculty Leave Approval or Rejection

option to view more information about a student by clicking the "Details" button. A responsive popup then opens, allowing faculty to examine the student's specific information more closely.

Overall, the FacultyPage streamlines administrative tasks, enabling faculty members to effectively manage leave requests and access pertinent student details.

4.5.6 Security Dashboard

The Security Page component (Figure 4.22) in React is designed to display a security guard's information along with the leave applications submitted by students. It utilizes axios to fetch security details and leave data from specified endpoints based on the Username obtained from the URL parameters. The component maintains various states to manage data, loading status, and potential errors.

Leave Applications							
Index	Roll No.	Reason	Date	Time Slot	Status	Actions	Details
1	160121737006	health issue	2023-11-14	12:15-01:00	Approved		Details
2	160121737002	Free Periods	2023-11-21	01:00-02:00	Approved		Details
3	160121737002	No classwork	2023-11-07	12:15-01:00	Approved		Details
4	160121737006	free Hours	2023-11-21	02:00-03:00	Pending	Approve Reject	Details
5	160121737021	sick	2023-11-21	10:10-11:10	Approved		Details
6	160121737021	free periods	2023-11-22	02:00-03:00	Approved		Details
7	160121737006	free periods	2023-11-22	12:15-01:00	Rejected		Details
8	160121737008	medical purpose	2023-11-25	11:10-12:15	Pending	Approve Reject	Details
9	160121737006	free periods	2023-11-28	02:00-03:00	Rejected		Details
10	160121737006	free hours	2023-11-29	02:00-03:00	Approved		Details
11	160121737006	free periods	2023-11-28	02:00-03:00	Rejected		Details
12	160121737006	sick	2023-11-29	02:00-03:00	Pending	Approve Reject	Details

Figure 4.21: Faculty Leave Requests Overview

It renders the security's personal information (such as username, email, and phone number) and lists student leave applications in a styled table format. For each leave application, the security can record whether the student has left the campus by marking their out-time verifying the identity of the student.

Security Information							
Email: cbitsecurity@gmail.com							
PhoneNumber: 9999999999							
UserName: CBITSecurity							

Leave Applications							
Index	Roll No.	Reason	Date	Time-Slot	Out-Time	Mark as Out	
1	160121737002	Free Periods	2023-11-21	01:00-02:00	Fri, 18 Apr, 2025, 10:11:46 pm		
2	160121737002	No classwork	2023-11-07	12:15-01:00	N/A	Camera No	
3	160121737006	free periods	2023-11-28	02:00-03:00	Sat, 19 Apr, 2025, 12:23:17 pm		
4	160121737021	free periods	2023-11-22	02:00-03:00	Sat, 19 Apr, 2025, 12:24:23 pm		
5	160121737006	health issue	2023-11-14	12:15-01:00	Sat, 19 Apr, 2025, 12:31:18 pm		

Figure 4.22: Security Dashboard

Before updating the out-time, a face recognition system is triggered to verify the student's identity. This system uses Tetrolet transforms to extract robust facial features for matching (Figure 4.23). Upon successful verification, the system updates the out-time on both the frontend and the backend server, ensuring accurate logging and enhancing campus security.

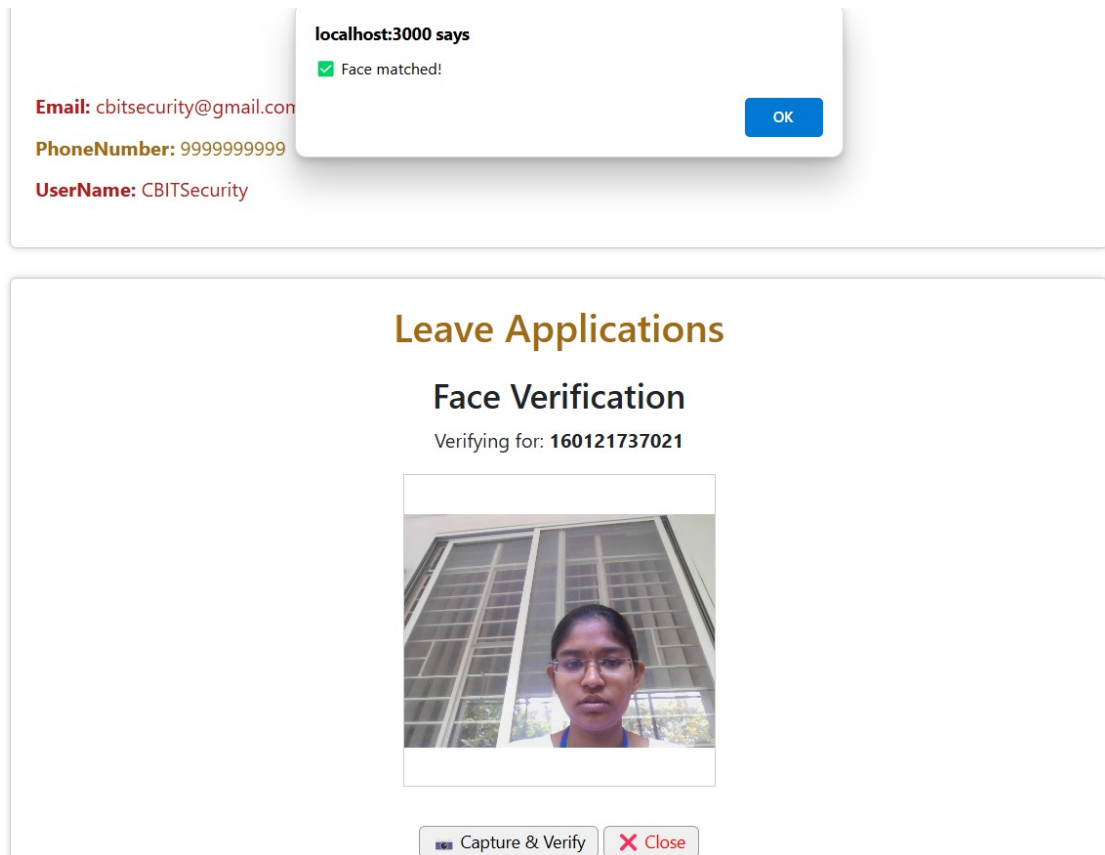


Figure 4.23: Face Verification using Tetrolet Transform-based Recognition

In conclusion, the user interfaces developed for the SMART EXIT system offer a seamless and intuitive experience tailored to the distinct roles of students, faculty, and security personnel. The design prioritizes clarity, functionality, and responsiveness, ensuring users can efficiently perform actions such as login, registration, leave application, and approval processes. Each interface is crafted with user-centric navigation and structured data presentation, enhancing accessibility and reducing operational friction. Together with the well-organized MongoDB-backed schema, these interfaces collectively establish a robust and scalable platform for academic leave management.

CHAPTER 5

Conclusion

This project delves into the use of wavelet-based neural networks for facial recognition, with a special focus on integrating Tetrolet transforms to improve feature extraction. Unlike traditional convolutional neural networks (CNNs), which process fixed-resolution pixel data, wavelet transforms provide a multiresolution approach that captures both spatial and frequency aspects of facial features. This capability makes the system exceptionally effective at addressing common challenges in face recognition, such as variations in lighting, changes in facial orientation, and occlusion.

The Tetrolet transform, an advanced and adaptive variant of the standard wavelet, takes this a step further by providing a more accurate representation of local structures within the face. It enhances the system's ability to capture finer details and subtle variations in facial features, leading to improved recognition performance. The Tetrolet transform also helps reduce the impact of noise and irrelevant data, ensuring that the system remains consistent and reliable even in diverse and challenging conditions.

The practical application of this technology spans real-time environments like surveillance, identity verification, campus monitoring, and access control systems. The system enables secure, automated identity verification, replacing traditional manual methods with more efficient, real-time authentication. The successful deployment of this system in real-world scenarios further validates its feasibility and effectiveness.

In conclusion, combining wavelet techniques, particularly the Tetrolet transform offers a powerful solution for advancing facial recognition technology. The proposed system delivers high accuracy, adaptability, and real-time performance, marking a significant leap forward in intelligent biometric systems.

CHAPTER 6

Future Scope

The future of wavelet-based facial recognition systems holds exciting prospects for further development and broader adoption. One key advancement is the extension of the current 2D recognition system to 3D face recognition, which will capture depth and structural details. This improvement will allow the system to maintain high performance even under challenging conditions such as varied lighting, different angles, and facial expressions. By applying wavelet transforms to volumetric data, the system will gain better spatial representation, enhancing both accuracy and resilience.

Another area for progress involves deploying the system on edge computing platforms like mobile phones, Raspberry Pi, and other embedded systems. This deployment ensures real-time face recognition with low latency, making it ideal for applications such as student attendance tracking, secure building access, smart surveillance, and home security. The compact yet powerful nature of the architecture ensures the models can function effectively, even in environments with limited resources.

To improve system reliability and inclusivity, future models may combine facial recognition with other biometrics, such as fingerprints or voice recognition. This multimodal approach will increase authentication accuracy and minimize false positives. In addition, to address privacy and ethical concerns, technologies like federated learning and encrypted computation can be employed, ensuring sensitive biometric data is processed securely. These features would also help meet data protection regulations, making the system more trustworthy and scalable for real-world applications.

In conclusion, this project sets the stage for creating secure, reliable, and scalable facial recognition systems that can be widely implemented in real-time applications across various sectors, including academia, public safety, smart cities, and more.

Bibliography

- [1] M. Saha, M. K. Naskar, and B. N. Chatterji, "Advanced Wavelet Transform for Image Processing—A Survey," in *Information, Photonics and Communication*, vol. 79, Springer, Singapore, 2020.
- [2] T. Guo, T. Zhang, E. Lim, M. López-Benítez, F. Ma, and L. Yu, "A review of wavelet analysis and its applications: Challenges and opportunities," *IEEE Access*, vol. 10, pp. 58869-58903, 2022, doi: 10.1109/ACCESS.2022.3179517.
- [3] A. Patil, R. Aggarwal, and A. Bahari, "UwU: Learnable Wavelet Unit with PR Constraint Relaxation for Enhanced CNN Feature Extraction," in *Proceedings of the 2023 IEEE International Conference on Visual Communications and Image Processing (VCIP)*, Jeju, Korea, Republic of, Dec. 2023, pp. 1–5, doi: 10.1109/VCIP59821.2023.10402772.
- [4] A. K. Alexandridis and A. D. Zapraniš, "Wavelet neural networks: A practical guide," *Neural Networks*, vol. 42, 2013, pp. 1–27. ISSN 0893-6080.
- [5] T. Wang, V. Guercio, P. Cattani, and F. Villecco, "A Review of Research Progress and Application of Wavelet Neural Networks," in *New Technologies, Development and Application VI. NT 2023*, vol. 687, Springer, Cham, 2023.
- [6] R. Xu, Y. Xu, X. Yang, H. Huang, Z. Lei, and Y. Quan, "Wavelet analysis model inspired convolutional neural networks for image denoising," *Applied Mathematical Modelling*, 2023.
- [7] Y. Pati and P. Krishnaprasad, "Analysis and synthesis of feedforward neural networks using discrete affine wavelet transforms," *IEEE Transactions on Neural Networks*, vol. 4, no. 1, pp. 73–85, 1993.
- [8] S. Mallat, "A Wavelet Tour of Signal Processing," Academic Press, 1999.

- [9] L. T. Antony and S. N. Kumar, “Demystifying the Applications of Wavelet Transform in Bio Signal and Medical Data Processing,” in *Smart Trends in Computing and Communications. SmartCom 2024*, vol. 945, Springer, Singapore, 2024.
- [10] I. Daubechies, “The wavelet transform, time-frequency localization and signal analysis,” *IEEE Transactions on Information Theory*, vol. 36, no. 5, pp. 961–1005, 1990.
- [11] A. Jamal, M. A. Hameed Ashour, R. A. Abbas Helmi, and S. L. Fong, ”A Wavelet-Neural Networks Model for Time Series,” *2021 IEEE 11th Symposium on Computer Applications & Industrial Electronics (ISCAIE)*, Penang, Malaysia, 2021, pp. 325–330.
- [12] M. Holschneider, “Wavelets: An Analysis Tool,” Springer, 1990.
- [13] Oscar Herrera, Belém Priego, David Pinto, Beatriz Beltrán, and Vivek Singh. ”Wavelets as activation functions in Neural Networks.” *Journal of Intelligent & Fuzzy Systems*, vol. 42, no. 5, 2022, pp. 4345–4355.
- [14] Q. Zhang and A. Benveniste, “Wavelet networks,” *IEEE Transactions on Neural Networks*, vol. 3, no. 6, pp. 889–898, 1992.
- [15] Szu, Harold, Telfer, Brian, and Kadambe, Shubha. “Neural network adaptive wavelets for signal representation and classification“ *Optical Engineering*, vol. 31, 1992, pp. 1907–1916.
- [16] P. Liang, C. Deng, J. Wu, and Z. Yang, ”Intelligent fault diagnosis of rotating machinery via wavelet transform, generative adversarial nets and convolutional neural network,” *Measurement*, 2020, article 107768.
- [17] Ramzi, Z., Michalewicz, K., Starck, J. L., et al. ”Wavelets in the Deep Learning Era.” *Journal of Mathematical Imaging and Vision*, vol. 65, 2023, pp. 240–251.
- [18] Y. Zhao, Y. Shen, Y. Zhu, and J. Yao, ”Forecasting Wavelet Transformed Time Series with Attentive Neural Networks,” *2018 IEEE International*

- Conference on Data Mining (ICDM)*, Singapore, 2018, pp. 1452–1457.doi: 10.1109/ICDM.2018.00201.
- [19] Y. Feng, T. Yang, and Y. Niu, "Subpixel Computer Vision Detection Based on Wavelet Transform," *IEEE Access*, vol. 8, pp. 88273-88281, 2020, doi: 10.1109/ACCESS.2020.2991846.
 - [20] Houda Meddeb, Zouhaira Abdellaoui, Firas Houaidi, "Development of surveillance robot based on face recognition using Raspberry-PI and IOT", *Microprocessors and Microsystems*, Vol. 96, 2023, 104728, ISSN 0141-9331,
 - [21] C. Tian, M. Zheng, W. Zuo, B. Zhang, Y. Zhang, and D. Zhang, "Multi-stage image denoising with the wavelet transform," *Pattern Recognition*, vol. 134, pp. 109050, 2023, doi: 10.1016/j.patcog.2022.109050.
 - [22] D. Cho and T. D. Bui, "Multivariate statistical modeling for image denoising using wavelet transforms," *Signal Processing: Image Communication*, vol. 20, no. 1, pp. 77-89, Jan. 2005, doi: 10.1016/j.image.2004.10.003.
 - [23] R. I. Bendjillali, M. Beladgham, K. Merit, and A. Taleb-Ahmed, "Improved Facial Expression Recognition Based on DWT Feature for Deep CNN," *Electronics*, vol. 8, no. 3, p. 324, Mar. 2019, doi: 10.3390/electronics8030324.
 - [24] W. Wang, X.-y. Sun, S. Karungaru, and K. Terada, "Face recognition algorithm using wavelet decomposition and Support Vector Machines," in *Proc. IEEE International Symposium on Optomechatronic Technologies (ISOT)*, Paris, France, 2012, pp. 1-6, doi: 10.1109/ISOT.2012.6403230.
 - [25] F. Tabassum, M. I. Islam, R. T. Khan, and M. R. Amin, "Human Face Recognition with Combination of DWT and Machine Learning," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 3, pp. 546-556, Mar. 2022, doi: 10.1016/j.jksuci.2020.02.002.

Identity Recognition Using Tetrolet Transforms In Surveillance Applications

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Abstract

Facial recognition has become a cornerstone technology in modern biometric systems, evolving significantly beyond traditional methods. This project explores the use of wavelet-based neural networks to enhance the accuracy and efficiency of face detection and recognition. Unlike conventional convolutional neural networks (CNNs) that focus on pixel-level image processing, wavelet transforms enable multiresolution analysis, capturing both spatial and frequency components of facial features. To further boost feature extraction, Tetrolet transforms—an advanced variant of wavelets—are employed for their ability to adaptively capture local geometric structures in facial images. This dual-level insight helps the model better handle variations in lighting, facial pose, and occlusion. By focusing on essential features and minimizing noise, the system achieves more reliable and robust performance. The integration of wavelet techniques with deep learning not only improves precision but also offers scalability for real-time applications such as security, surveillance, and identity verification. The project demonstrates the advantages of Tetrolet-based feature extraction over traditional methods, offering a flexible and high-performing solution for face recognition challenges.

Keywords: Tetrolet Transforms, Facial Recognition, Wavelet Neural Networks, Multiresolution Analysis

1 Introduction

The evolution of image processing technologies has significantly enhanced various applications, such as face detection, biometric systems, and human-computer interaction. Central to these advancements is the ability to extract meaningful features from images, which directly impacts the performance and accuracy of subsequent tasks like classification and recognition. Traditional methods like Haar wavelet transforms have long been used for face detection due to their ability to decompose images into different frequency components. However, as the complexity of modern image data continues to increase, there is a growing need for more advanced methods that can handle the fine-grained spatial and directional features inherent in facial structures. This paper presents a novel approach to face detection by combining wavelet-based techniques—specifically Haar wavelets, O-shape transformations, T-shape transformations, I-shape transformations and L-Shape transforms—into a unified framework designed to improve the accuracy and adaptability of image feature extraction.

Wavelet transforms, particularly the Haar transform, have been widely used in image processing because of their simplicity and effectiveness in decomposing images into multiple resolutions. Haar transforms work by averaging and differencing pixel values within a region, which helps capture both the low-frequency background and high-frequency details such as edges. However, Haar wavelets are limited in their ability to capture directional features, which are particularly important in facial feature extraction. Faces contain complex structures, and traditional methods may not be effective at distinguishing subtle features like the curvature of the eyes or the shape of the nose.

To address this, we propose enhancing Haar-based techniques with O-shape and T-shape transformations. The O-shape transformation modifies the standard block-wise wavelet decomposition by applying it to localized 4x4 blocks, preserving high-frequency features such as edges around critical facial regions. This approach allows for better representation of features that are essential for accurate face detection, particularly in the areas of the eyes, nose, and mouth. The T-shape transformation, on the other hand, restructures the Haar-based decomposition to focus on T-shaped regions within the image blocks. By emphasizing directional features, this transformation captures the spatial orientation of facial features that are often overlooked by traditional wavelet methods.

2 Literature Review

Wavelet Neural Networks (WNNs) have emerged as a promising approach to improving feature extraction and function approximation in various domains, including signal processing, image denoising, time-series forecasting, and face recognition. The integration of wavelet transforms with neural networks has been explored extensively to enhance model performance and provide better localization in both time and frequency domains. This survey summarizes key research works related to WNNs, highlighting their contributions and limitations.

2.0.1 S. Mallat, “A Wavelet Tour of Signal Processing” (1999)

Mallat’s work is instrumental in providing theoretical underpinnings for the use of wavelets in neural networks, which are leveraged for capturing both time and frequency information. In *A Wavelet Tour of Signal Processing*, Stéphane Mallat presents an in-depth exploration of wavelet theory and its application in signal processing. The book outlines fundamental concepts of wavelet transforms, sparse representations, and computational harmonic analysis, offering an extensive foundation in both the mathematical theory and practical implementations of wavelets. Covered topics, the Fourier analysis, time-frequency dictionaries, and approximation methods. Mallat delves into specific techniques for sampling, compression, and denoising, along with applications of wavelet transforms in image processing, edge detection, and multiresolution analysis. Contributions from Gabriel Peyré enrich the content, providing insights into modern signal processing techniques and the role of wavelets in sparse representations.

2.0.2 L. T. Antony and S. N. Kumar, “Demystifying the Applications of Wavelet Transform in Bio Signal and Medical Data Processing” (2024)

This paper focuses on the use of wavelet transforms in bio-signal analysis, emphasizing the flexibility of wavelets in medical data processing, where handling complex, multidimensional data is crucial. The paper, *Demystifying the Applications of Wavelet Transform in Bio Signal and Medical Data Processing* by Linu Tess Antony and S. N. Kumar, presented at SmartCom 2024, discusses the transformative role of wavelet transforms in bio-signal and medical imaging. It focuses on discrete wavelet transform (DWT) and wavelet packet transform (WPT), which enable signal and image analysis by providing a localized representation in time and frequency domains. These wavelet techniques are invaluable for denoising, feature extraction, and classification of bio-signals like ECG and EEG. In medical imaging, wavelet transforms enhance denoising, compression, fusion, and segmentation, contributing to more accurate diagnoses and supporting medical advancements.

2.0.3 I. Daubechies, “The Wavelet Transform, Time-Frequency Localization, and Signal Analysis” (1990)

Daubechies’ work presents the mathematical basis for wavelet transforms, showing how they enable precise time-frequency localization, which is especially valuable in applications involving non-stationary signals. In “The Wavelet Transform, Time-Frequency Localization, and Signal Analysis,” published in *IEEE Transactions on Information Theory*, Ingrid Daubechies compares two approaches for analyzing time-dependent signals locally in time: the short-time Fourier transform (STFT) and the wavelet transform. The wavelet method offers enhanced time resolution for high-frequency components compared to low-frequency ones. Daubechies thoroughly examines the reconstruction processes and stability of both techniques in relation to time-frequency density. This paper formalizes the concept of time-frequency localization through theorems, providing a foundation for understanding the unique advantages of wavelets in signal analysis.

2.0.4 A. Jamal, M. A. Hameed Ashour, R. A. Abbas Helmi, and S. L. Fong, “A Wavelet-Neural Networks Model for Time Series” (2021)

This paper introduces a model that combines wavelets with neural networks for enhanced time-series forecasting, demonstrating improved prediction accuracy compared to traditional models. In the 2021 paper A Wavelet-Neural Networks Model for Time Series, Arshad Jamal and colleagues present a hybrid forecasting model that integrates Wavelet Transforms (WT) with Artificial Neural Networks (ANN) to improve time-series prediction accuracy. This model is proposed as an alternative to the traditional ARIMA method, aiming to minimize forecasting errors significantly. By utilizing WT to select inputs for the ANN, as opposed to the conventional ARIMA-based input selection, the model achieves notably lower errors, as measured by metrics like MSE, RMSE, and MAPE. Experimental results show an error reduction nearing 99.92%, demonstrating the model’s effectiveness and efficiency in processing time-series data for enhanced predictive performance.

2.0.5 M. Holschneider, “Wavelets: An Analysis Tool” (1990)

Holschneider’s book delves into wavelet-based analysis tools that can reduce data complexity effectively, a critical advantage in fields requiring data compression and pattern recognition. In Wavelets: An Analysis Tool (2023), M. Holschneider provides a comprehensive introduction to wavelet analysis, exploring its wide-ranging applications in signal data analysis, such as geoprospection, speech recognition, and singularity detection, as well as in data compression for image and audio signals. The book is structured to be accessible, starting with a detailed explanation of the continuous wavelet transform and progressing to the discretized version, or ortho-normal basis. Holschneider covers topics like wavelet analysis of regular functions, tempered distributions, and square integrable functions. The text emphasizes a group theory approach to unify different wavelet methods, making it a valuable resource for students and professionals in fields like mathematics, physics, and signal processing.

2.0.6 M. Saha et al., “Advanced Wavelet Transform for Image Processing—A Survey” (2020)

This survey explores the use of advanced wavelet transforms in image processing, which are instrumental in enhancing image details and resolution, showing potential in high-precision image recognition. The survey paper Advanced Wavelet Transform for Image Processing by Manas Saha, Mrinal Kanti Naskar, and B. N. Chatterji (2019) examines the evolving role of wavelet transform in multiresolution image analysis. This work discusses ten advanced wavelet techniques, addressing their importance, conceptual foundations, structural details, benefits, and limitations within various image processing applications. The authors analyze relationships among these advanced wavelets and provide a comparison table that highlights current trends and applications, offering insights into recent developments in wavelet-based image processing techniques.

2.0.7 T. Guo et al., “A Review of Wavelet Analysis and Its Applications: Challenges and Opportunities” (2022)

This review covers both applications and challenges in wavelet analysis, addressing scalability and real-time processing limitations, with insights relevant for improving WNNs in large-scale data processing. The paper *A Review of Wavelet Analysis and Its Applications: Challenges and Opportunities* by Tiantian Guo, Miguel López-Benítez, Tongpo Zhang, Enggee Lim, Fei Ma, and Limin Yu presents an extensive overview of wavelet theory, highlighting its evolution, applications, and integration with neural networks. The authors examine wavelet theory’s foundational construction and development, emphasizing key models, algorithms, and the advantages of Rational Wavelet Transform (RWT) over conventional methods. A significant portion of the review focuses on Wavelet Neural Networks (WNN), discussing their system design, implementation, and potential for future applications when combined with other neural network algorithms. The review also provides a comparative analysis of wavelet-based applications, summarizing wavelet analysis’s benefits across various scenarios. The authors conclude by identifying emerging research challenges, paving the way for innovative wavelet-based applications and system architectures.

2.0.8 Y. Pati and P. Krishnaprasad, “Analysis and Synthesis of Feedforward Neural Networks using Discrete Affine Wavelet Transforms” (1993)

This study integrates wavelet transforms with feedforward neural networks, highlighting structural benefits that can be applied to WNN architectures. The paper *Analysis and Synthesis of Feedforward Neural Networks Using Discrete Affine Wavelet Transformations* by Y.C. Pati and P.S. Krishnaprasad explores how feedforward neural networks with sigmoidal activation functions can be represented through discrete affine wavelet transformations. This representation enables the analysis and synthesis of these networks by interpreting them as systems that perform affine wavelet decompositions on mappings. The wavelet transform framework provides a mathematical foundation to analyze the kinds of mappings these networks can implement and how to precisely achieve such mappings. Additionally, the authors leverage the spatial and spectral localization characteristics of wavelets to develop two synthesis methods that reduce network training to convex optimization, streamlining the process of designing neural networks for specific approximation tasks.

2.0.9 O. Herrera et al., “Wavelets as Activation Functions in Neural Networks” (2022)

This paper explores the novel use of wavelets as activation functions in neural networks, which can increase responsiveness to varied data types and suggests enhancements for WNN adaptability. The article *Wavelets as Activation Functions in Neural Networks* by Oscar Herrera and Belém Priego, published in the *Journal of Intelligent & Fuzzy Systems*, explores the use of wavelets as activation functions in neural networks. Traditional activation functions, such as threshold, sigmoid, hyperbolic tangent, ReLU, GELU, and Soft-plus, have dominated neural network architectures, but the authors

address the ongoing research to identify novel activation functions that can improve model performance. This study specifically examines Gaussian family wavelets (including their first, second, and third derivatives) as potential activation functions within neural networks. Results from experiments using Keras-TensorFlow demonstrate that these wavelet-based activation functions can enhance network performance, supporting the potential for wavelets to be integrated as standard options on high-performance deep learning platforms.

2.0.10 A. D. Le, S. Jin, Y. S. Bae and T. Nguyen, "A Novel Learnable Orthogonal Wavelet Unit Neural Network with Perfection Re construction Constraint Relaxation for Image Classification" (2023)

The paper "A Novel Learnable Orthogonal Wavelet Unit Neural Network with Perfection Reconstruction Constraint Relaxation for Image Classification" by An D Le, Shiwei Jin, You Suk Bae, and Truong Nguyen, presented at the 2023 IEEE International Conference on Visual Communications and Image Processing (VCIP), introduces the UwU unit, a learnable wavelet-based unit designed to enhance convolutional neural networks (CNNs). This approach addresses the limitations of CNNs, which typically use lowpass frequency features through max pooling operations. The authors propose using a one-layer fully connected network (FCN) incorporating both coarse and detailed components via Discrete Wavelet Transform (DWT), with the key innovation being the trainable coefficients of the UwU unit and its incorporation of Perfection Reconstruction (PR) constraint relaxation for feature map fine-tuning. The paper shows that integrating UwU into ResNet architectures leads to significant performance improvements, especially on complex datasets like ImageNet1K and DTD, and even achieves competitive anomaly detection results in the MVTecAD dataset when integrated into a ResNet18-based encoder for the CFLOW-AD system.

2.0.11 A. K. Alexandridis and A. D. Zaprani, "Wavelet Neural Networks: A Practical Guide" (2013)

This guide discusses practical methods for implementing WNNs, covering common applications and technical challenges. The paper "Wavelet Neural Networks: A Practical Guide" by Antonios K. Alexandridis and Achilleas D. Zaprani presents a comprehensive framework for applying Wavelet Networks (WNNs) across various applications. While WNNs have shown great success in many fields, a universally accepted framework for their application was previously lacking. The authors propose a statistical model identification framework that covers key areas such as the structure of WNNs, training methods, initialization algorithms, variable significance and selection algorithms, model selection techniques, and the construction of confidence and prediction intervals. They also discuss the complexity of these algorithms. The proposed framework was tested on two simulated cases and three real-world datasets, including chaotic time series from the Mackey-Glass equation, daily temperatures in Berlin, daily wind speeds in New York, and breast cancer classification. The results demonstrated

that the framework yields stable and robust outcomes, highlighting its applicability across different domains.

2.0.12 T. Wang et al., “A Review of Research Progress and Application of Wavelet Neural Networks” (2023)

This comprehensive review discusses recent advancements in WNNs and emphasizes the potential of hybrid WNNs for overcoming current limitations. The paper “A Review of Research Progress and Application of Wavelet Neural Networks” by Tonghao Wang, Vincenzo Guercio, Piercarlo Cattani, and Francesco Villecco, published in *New Technologies, Development and Application VI* (2023), provides a comprehensive review of the development and applications of Wavelet Neural Networks (WNNs). The combination of wavelet transform theory with artificial neural networks (ANNs) has led to the creation of WNNs, a specialized network structure that optimizes neural network performance. This paper summarizes the evolution of WNNs, including their system structure, algorithm implementation, and derivative models, and explores cutting-edge applications in various fields. The authors highlight that integrating wavelet theory with neural networks enhances the speed of convergence and the accuracy of models. WNNs are rapidly advancing in applications such as audio signal and image processing, making them a promising area for future research and innovation. This work aims to provide a foundation for further studies on WNN-based applications and new network model designs.

2.0.13 R. Xu et al., “Wavelet Analysis Model Inspired Convolutional Neural Networks for Image Denoising” (2023)

This paper addresses the trade-off between computational efficiency and denoising performance in convolutional neural network (CNN)-based image denoising methods. While CNNs have shown continuous improvement in performance, their computational cost has increased due to complex network architectures. To strike a balance, the authors propose a CNN model inspired by traditional variational models with wavelet analysis operators. The proposed model includes four key modules: iterative encoding-decoding units, inspired by the iterative denoising process; directional convolutions, based on separable wavelet filters; inception modules, reflecting multi-scale wavelet analysis; and stage-wise connections, inspired by the process of adding noise back. The model is designed to improve both denoising performance and computational efficiency. Experimental results demonstrate that the proposed CNN offers competitive denoising performance while maintaining high computational efficiency during both training and testing, outperforming existing state-of-the-art methods.

2.0.14 Q. Zhang and A. Benveniste, “Wavelet Networks” (1992)

This paper proposes a wavelet network concept as an alternative to traditional feed-forward neural networks for approximating arbitrary nonlinear functions. The core idea is to replace the conventional neurons with “wavelons,” which are computational units formed by combining an affine transform with a multidimensional wavelet. These wavelons are capable of efficiently representing complex nonlinear functions.

The paper further addresses the challenge of identifying the affine transforms and synaptic weights from potentially noisy input/output data. To train the wavelet network, the authors introduce a backpropagation-type algorithm specifically designed for this purpose. Experimental results are provided to demonstrate the effectiveness of the wavelet network in function approximation tasks, showcasing its potential advantages over traditional neural network approaches. This approach aims to offer a more flexible and accurate method for solving nonlinear problems, especially when dealing with noisy data.

2.0.15 H. Szu et al., “Neural Network Adaptive Wavelets for Signal Representation and Classification” (1992)

This paper presents methods for adaptively generating wavelet templates for signal representation and classification using neural networks. The authors introduce the concept of a “super-wavelet,” which is a linear combination of wavelets treated as a single wavelet. This super-wavelet allows the shape of the wavelet to be adjusted to better suit specific problems, offering a more flexible approach than simply adjusting the parameters of a fixed-shape wavelet. The paper outlines the necessary network structures and energy functions for both signal representation and classification tasks. Simulations are provided for one-dimensional signals, but the authors note that the approach can be extended to image processing as well. Additionally, the paper discusses potential applications of this adaptive wavelet framework in phoneme and speaker recognition, suggesting that these methods could be highly effective for audio signal analysis. Overall, the paper emphasizes the adaptability of the super-wavelet and its potential for improving signal processing tasks in various domains, highlighting its advantages over traditional wavelet approaches.

2.0.16 P. Liang et al., “Intelligent Fault Diagnosis of Rotating Machinery via Wavelet Transform, GAN, and CNN” (2020)

This paper presents a novel method for diagnosing faults in rolling bearings, which are crucial components in rotating machinery. Traditional fault diagnosis techniques typically rely on complex, manually crafted feature extraction, requiring significant human expertise. Emerging deep learning methods can reduce this dependence but often require large datasets of fault signals, which are difficult to obtain in real-world applications. To address these challenges, the authors propose a hybrid approach combining Convolutional Neural Networks (CNN) and Support Vector Machines (SVM). The Continuous Wavelet Transform (CWT) is applied to convert one-dimensional vibration signals into two-dimensional time-frequency images. These images are then used to train the model followed by diagnosing the fault’s location and severity. The method is tested on two benchmark datasets: the CWRU and MFPT datasets. The experimental results show that the proposed approach outperforms other advanced diagnostic techniques, offering higher accuracy and stability in fault detection.

2.0.17 Z. Ramzi et al., “Wavelets in the Deep Learning Era” (2023)

This paper presents a comprehensive framework for the design and analysis of U-Net architectures, which are widely used for tasks involving continuous signals on a square. The authors explore the roles of the encoder and decoder in U-Nets, examining their scaling limits and their relationship to ResNets through preconditioning. They introduce Multi-ResNets, a simplified version of U-Nets that uses a wavelet-based encoder without learnable parameters. Additionally, the paper discusses how novel U-Net architectures can be designed to incorporate function constraints, natural bases, or data geometry. The framework is applied to diffusion models, where it identifies that high-frequency information decays rapidly due to noise, and demonstrates how U-Nets with average pooling leverage this property. Through experiments, the authors show that Multi-ResNets outperform classical U-Nets in tasks such as image segmentation, PDE surrogate modeling, and generative modeling with diffusion models. The paper highlights the potential of the proposed framework to improve the theoretical understanding and design of U-Nets, opening up new possibilities for scalable neural architectures in a variety of applications beyond traditional square-shaped data.

2.0.18 Y. Zhao et al., “Forecasting Wavelet Transformed Time Series with Attentive Neural Networks” (2018)

This paper presents an innovative approach to flight trajectory prediction, focusing on improving accuracy in air traffic control, particularly during maneuver operations. Traditional data-driven methods typically frame this task as time series forecasting, which often struggles to maintain high accuracy. The paper highlights the underutilization of frequency-domain analysis, a key method in time series forecasting, for this problem. To address this, the authors propose a wavelet transform-based framework that performs time-frequency analysis of flight patterns to enhance trajectory forecasting. The method involves an encoder-decoder neural network architecture designed to estimate wavelet components, which effectively model both global flight trends and local motion details. The framework is validated using a real-world dataset, and experimental results show superior accuracy compared to existing baselines. The proposed approach notably improves prediction performance during the climb and descent phases, where maneuver control is critical. The study confirms that time-frequency analysis is an effective technique for flight trajectory prediction, offering enhanced precision in real-world applications of air traffic management.

2.0.19 Y. Feng et al., “Subpixel Computer Vision Detection Based on Wavelet Transform” (2020)

The paper “Subpixel Computer Vision Detection Based on Wavelet Transform” by Yuye Feng, Ting Yang, and Yongfeng Niu, published in IEEE Access (Volume 8), explores advancements in computer vision detection using wavelet transforms for enhanced accuracy. Traditional computer vision detection techniques often lack the precision required in applications such as transportation, public security, and defense. This study proposes a subpixel detection algorithm that integrates a quantum denoising approach using dual-tree dual-density wavelet transforms, which allows for precise

image representation and effective noise reduction. Additionally, the method incorporates cubic spline interpolation edge detection to pinpoint image edges at subpixel accuracy by identifying zero crossings in a second-order function of the cubic spline. Comparisons with traditional methods demonstrate the proposed algorithm's robustness, improved computational speed, and adaptability, marking it as a promising solution for high-precision applications.

**2.0.20 Houda Meddeb, Zouhaira Abdellaoui, Firas Houaidi,
"Development of surveillance robot based on face
recognition using Raspberry-PI and IOT",2023**

In the paper "Development of Surveillance Robot Based on Face Recognition Using Raspberry-PI and IoT" by Houda Meddeb et al., a low-cost surveillance robot system uses Haar Cascade Classifier and LBPH for facial recognition. However, these methods can struggle with complex environments and varying lighting conditions, often leading to limitations in recognition accuracy and increased false positives. To address this, we propose integrating wavelet neural networks (WNNs), which offer multi-resolution analysis by capturing high and low-frequency image features, improving facial recognition under challenging conditions. WNNs would enhance feature extraction and provide more accurate identity recognition, benefiting the robot's performance in security-sensitive applications. However, this integration introduces additional computational demands on the Raspberry Pi, potentially requiring processing optimizations or edge-computing support to maintain real-time performance in IoT contexts. Despite these challenges, WNNs could significantly increase robustness and adaptability, making the system more reliable for diverse, high-security scenarios.

**2.0.21 Chunwei Tian, Menghua Zheng, Wangmeng Zuo, Bob
Zhang, Yanning Zhang, David Zhang, "Multi-stage image
denoising with the wavelet transform," 2023**

The paper "Multi-stage Image Denoising with the Wavelet Transform" by Chunwei Tian et al., published in Pattern Recognition (Volume 134, February 2023), presents a novel approach for image denoising using a multi-stage convolutional neural network (CNN) integrated with wavelet transform. The proposed method, called MWDCNN, involves three main stages: a dynamic convolutional block (DCB), two cascaded wavelet transform and enhancement blocks (WEBs), and a residual block (RB). The DCB uses dynamic convolution to adjust parameters for balancing denoising performance with computational efficiency. The WEB blocks combine wavelet transforms and discriminative learning to effectively suppress noise while preserving fine image details. The RB further refines the features to remove redundant information, improving the denoising outcome and producing cleaner images using enhanced residual dense architectures. Experimental results demonstrate that MWDCNN outperforms other popular denoising methods both qualitatively and quantitatively, making it a powerful solution for image denoising tasks.

2.0.22 Dongwook Cho, Tien D. Bui, "Multivariate statistical modeling for image denoising using wavelet transforms," 2005

The paper "Multivariate Statistical Modeling for Image Denoising Using Wavelet Transforms" by Dongwook Cho and Tien D. Bui, introduces a new approach for image denoising by leveraging multivariate statistical theory in the wavelets. The authors derive an estimation rule to obtain denoised coefficients from a noisy image, using a parametric multivariate generalized Gaussian distribution (MGGD) model that closely fits the empirical distribution of the original clean image. This multivariate model accounts for dependencies between wavelet coefficients, including relationships between coefficients in different subbands and neighboring coefficients. The proposed method is shown to outperform existing denoising methods based on wavelets, achieving the highest peak signal-to-noise ratio (PSNR) among methods that use the same wavelet filter (Daubechies 8), demonstrating its effectiveness in image denoising.

2.0.23 Ridha Ilyas Bendjillali, Mohammed Beladgham, Khaled Merit, Abdelmalik Taleb-Ahmed, "Improved Facial Expression Recognition Based on DWT Feature for Deep CNN," 2019

This paper introduces a method for facial expression recognition (FER) that combines the Viola–Jones face detection algorithm, contrast limited adaptive histogram equalization (CLAHE), discrete wavelet transform (DWT), and deep convolutional neural networks (CNN). The Viola–Jones algorithm is first used to detect faces and facial parts, followed by image enhancement using CLAHE. DWT is then applied to extract facial features, which are fed into a CNN for classification. The method was evaluated on the CK+ and JAFFE face databases, achieving recognition accuracies of 96.46% and 98.43%, respectively. The approach proves robust against illumination changes.

2.0.24 Wei Wang, Xiang-yu Sun, Stephen Karungaru, Kenji Terada, "Face recognition algorithm using wavelet decomposition and Support Vector Machines," 2012

This paper presents a face recognition algorithm that combines wavelet decomposition for feature extraction and Support Vector Machines (SVM) for classification. The proposed method improves recognition precision by using wavelet decomposition, which is less sensitive to variations in facial expressions. The face region is first detected using an improved AdaBoost algorithm, and appropriate features are extracted through wavelet decomposition. These features are then used to train the SVM model with three different kernel functions: Radial basis function, Polynomial, and Linear kernel. Experimental results on the Ren-FEdb database show a recognition accuracy of 96.78

2.0.25 Fahima Tabassum, Md. Imdadul Islam, Risala Tasin Khan, M.R. Amin, "Human face recognition with combination of DWT and machine learning," 2022

This paper presents a novel approach to human face recognition by combining Discrete Wavelet Transform (DWT) with four different algorithms: error vector of Principal Component Analysis (PCA), eigen vector of PCA, eigen vector of Linear Discriminant Analysis (LDA), and Convolutional Neural Networks (CNN). The results of these algorithms are combined using entropy of detection probability and a fuzzy system. The approach demonstrated an accuracy rate of 89.56% in the worst case and 93.34% in the best case, surpassing previous methods that applied individual techniques to specific datasets. The paper concludes that recognition accuracy depends on the image quality and diversity of the database.

Methodology

3 Tetrolet Transformations

3.1 Introduction to Tetrolet Transformations

Tetrolet transformations extend the classical Haar wavelet transform by incorporating adaptive tetromino-based tiling strategies for localized signal and image processing. This method offers superior flexibility in analyzing image structures, particularly in applications where geometric adaptability is crucial. Unlike traditional wavelet transformations that rely on uniform partitions, the tetrolet approach enables dynamic segmentation, enhancing the capability to capture structural patterns within data.

In order to achieve an efficient image representation, we employ a new adaptive Haar wavelet transform, called the Tetrolet Transform. Tetrolets are Haar-type wavelets whose supports are tetrominoes—shapes formed by connecting four equal-sized squares. The corresponding fast filter bank algorithm is simple yet highly effective. At every level of the filter bank algorithm, the low-pass image is divided into 4×4 blocks. Within each block, a local tetrolet basis is determined, adapting to the image geometry. This adaptive segmentation enables efficient signal compression and feature extraction.

Tetrominoes, popularized through the classic computer game 'Tetris,' are formed from a union of four unit squares, each connected by edges rather than merely at their corners. Disregarding rotations and reflections, five unique tetromino shapes exist, providing a structured yet flexible basis for adaptive wavelet transformations. This approach significantly enhances the multi-resolution analysis capability of wavelet-based methods.

3.2 Mathematical Foundation of Tetrolet Transformations

The tetrolet transform builds upon the classical Haar wavelet, extending it through adaptive basis selection. Unlike standard wavelets that operate on fixed rectangular or square partitions, the tetrolet transform dynamically selects tetromino shapes, adjusting to the image's local features. This adaptability enhances its performance in image compression, denoising, and edge detection tasks.

In this section, we explore the mathematical formulation of the Haar wavelet transform, which is applied to image data for multi-resolution analysis. The Haar wavelet transform works by dividing the image into smaller blocks and performing a series of operations to extract various frequency components. This section outlines the decomposition of the image into different components using pixel values, with an emphasis on understanding the calculation of Haar coefficients.

The Haar wavelet transform operates on a 2x2 block of pixels from an image. Let the pixel values in the 2x2 block be represented by $I_{i,j}$, where i and j represent the row and column indices of the block. The Haar transform computes four main components: the approximation coefficient, horizontal detail, vertical detail, and diagonal detail.

Approximation Coefficient

The approximation coefficient A is calculated as the average of all pixel values in the 2x2 block, representing the low-frequency content of the image. The formula for the approximation coefficient is given by:

$$A = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 I_{i,j} \quad (1)$$

Horizontal Detail Coefficient

The horizontal detail coefficient H is used to capture the horizontal edges in the image. It is calculated by alternating the sign of the pixel values along each row (i.e., along the columns), based on the column index j :

$$H = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^j I_{i,j} \quad (2)$$

Vertical Detail Coefficient

The vertical detail coefficient V highlights the vertical edges in the image. It is computed by alternating the sign of the pixel values along each column (i.e., along the rows), based on the row index i :

$$V = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^i I_{i,j} \quad (3)$$

Diagonal Detail Coefficient

The diagonal detail coefficient D is responsible for capturing the diagonal edges in the image. This coefficient is calculated by alternating the sign of the pixel values both row-wise and column-wise, based on the sum of the indices $i + j$:

$$D = \frac{1}{4} \sum_{i=1}^2 \sum_{j=1}^2 (-1)^{i+j} I_{i,j} \quad (4)$$

Significance of the $(-1)^{i+j}$ Pattern

The term $(-1)^{i+j}$ plays a key role in the Haar wavelet transform. It alternates the signs of the pixel values, which helps in distinguishing different frequency components. Specifically:

- If $i + j$ is **even**, $(-1)^{i+j} = +1$, meaning the pixel value remains unchanged.
- If $i + j$ is **odd**, $(-1)^{i+j} = -1$, meaning the pixel value is inverted.

This alternating sign pattern enables the Haar transform to effectively capture horizontal, vertical, and diagonal edges in the image.

3.3 Graphical Representation of Tetromino Tiling

To illustrate the concept, Figure 2 showcases an example of tetromino-based tiling used in tetrolet transformations.

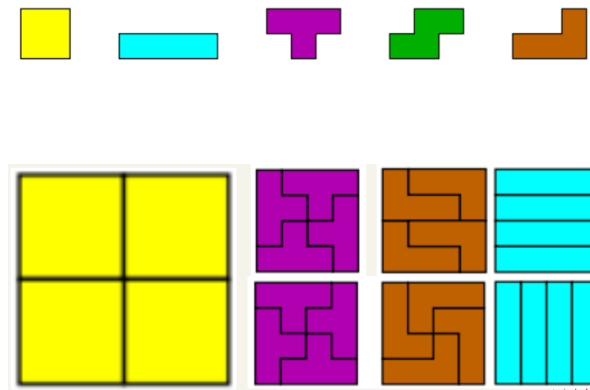


Fig. 1: Tetromino Tiling Patterns in Tetrolet Transformation

3.4 Applications and Future Work

Tetrolet transforms are highly beneficial in image compression, denoising, and pattern recognition tasks. Future research aims to integrate tetrolet-based neural networks for deep learning applications, leveraging their geometric adaptability for feature extraction in convolutional architectures. The fusion of tetrolet transformations with emerging AI techniques could further enhance real-time image processing capabilities.

4 Methodology

The methodology for this face image classification problem involves several key steps, starting from image preprocessing to feature extraction and model training. The process can be broken down into the following stages:

4.1 Dataset Description

The dataset used for this face image classification project is derived from a specialized Indian facial image database that addresses the unique challenges posed by individuals wearing spectacles. Spectacles often cause several issues, such as occlusion, glare, and reflections, which hinder accurate detection and recognition of facial features, especially the eyes. These challenges are collectively termed the "spectacle problem."

4.1.1 Spectacle Problem

The spectacle problem includes a range of issues that interfere with accurate face and eye detection in images. These problems typically arise from:

- **Occlusion:** Parts of the face, especially the eyes, may be partially or fully covered by the spectacles' frame, making it difficult for detection algorithms to identify key features.
- **Glare:** Light reflections on the lenses of spectacles can create bright spots on the image, obscuring important facial features like the eyes and nose.
- **Secondary Reflection:** Reflections from the lenses themselves can distort the appearance of the eyes and face, complicating accurate recognition.

These problems are significant in real-world applications where individuals may frequently wear spectacles, and conventional face detection systems struggle to handle such scenarios effectively.

4.1.2 Indian Facial Image Database

To address these challenges, the Indian facial image database was specifically created to focus on facial images with and without spectacle-related problems. The database is divided into two versions:

- **Version 1:** The first release of the dataset contains 271 images that showcase various spectacle-related issues, such as glare and occlusion. These images help train models that can handle the specific problems associated with spectacles.
- **Version 2:** The second release significantly expands the dataset to 1,23,213 images collected from 58 subjects. This version offers a more comprehensive collection of images, representing diverse facial structures and lighting conditions. The dataset also includes images with various spectacle-related problems like glare, occlusion, and secondary reflections.

4.1.3 Metadata and Nomenclature

The dataset is organized using a systematic nomenclature format to categorize each image based on the following parameters:

- **NUM:** Subject ID Number (from 000 to 270 for Version 1 and a broader range for Version 2).
- **G:** Gender of the subject (Female = 0, Male = 1).
- **S:** Spectacles (Without spectacles= 0, With spectacles=1).

- **I:** Illumination Problem (No illumination problem=0, illumination problem=1).
- **O:** Occlusion (No occlusion=0, With occlusion = 1).

The database is provided in various formats, including ZIP archives containing image files (in JPG, BMP, and TXT formats), which can be accessed for training and testing purposes. The images are accompanied by their corresponding metadata to assist in understanding and handling the specific challenges posed by spectacles during face recognition tasks.

4.1.4 Data Access and License

The Indian facial image database is publicly available under the Creative Commons Attribution License. The license allows users to freely use and share the data while giving appropriate credit to the creators of the dataset.

4.1.5 Relevance to the Project

This dataset is highly relevant to the current face image classification task, as it provides a realistic set of images with the types of occlusion, glare, and reflections typically encountered in real-world scenarios. The use of this database allows for better handling of spectacle problems, which is essential for improving face and eye detection accuracy, particularly in cases where spectacles obscure key facial features.

4.2 Preprocessing

The following preprocessing steps are applied to the face images:

- **Grayscale Conversion:** Each image is converted into grayscale to reduce the complexity and focus on structural features.
- **Resizing:** All images are resized to 256×256 pixels to standardize input sizes.

4.3 Image Block Division

Once the images are preprocessed, they are divided into non-overlapping 4×4 blocks. This allows for local analysis and transformation of each smaller region, enabling the extraction of detailed information from different parts of the image.

4.4 Tetrolet Transform

The Tetrolet transform is a block-based transformation technique designed to capture local geometric structures in images more effectively than traditional wavelets. It begins by dividing the image into non-overlapping 4×4 pixel blocks. Within each block, the transform rearranges pixels based on optimal tetromino tiling patterns, which are inspired by the game Tetris. This process allows for adaptive partitioning of the image content according to its local structure, enabling the transform to align more closely with edges and contours present in the image.

By adaptively selecting the best tiling configuration that minimizes approximation error, the Tetrolet transform provides superior energy compaction and sparse representation compared to fixed wavelet bases. This step is particularly useful for

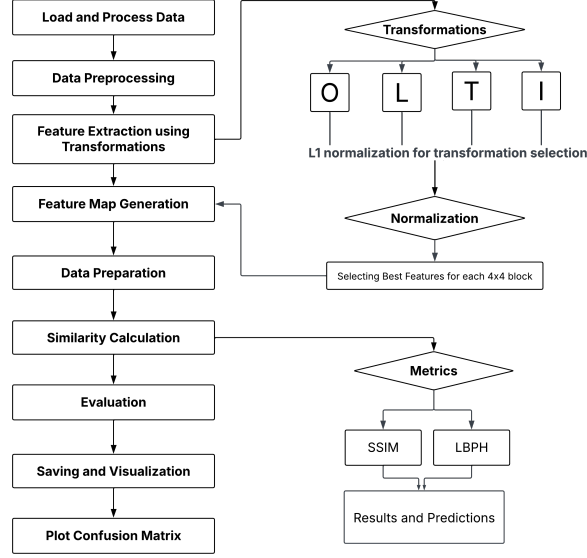


Fig. 2: Methodology flowchart

preserving local features such as corners, edges, and textures. The rearranged blocks are then subjected to further transformations, such as quantization or compression, in subsequent stages of processing. As a result, the Tetrolet transform serves as an effective pre-processing tool that enhances image representation, making it well-suited for applications in image compression, denoising, and pattern recognition.

4.5 Sub-block Division and Haar Wavelet Transformation

Each 4×4 block is further divided into four 2×2 sub-blocks. For each of these 2×2 sub-blocks, the Haar wavelet transform is applied. The Haar wavelet transformation computes the following coefficients:

- **A (Approximation):** The average of the pixels within the block.
- **H (Horizontal Detail):** The difference between the pixel values along each row.
- **V (Vertical Detail):** The difference between the pixel values along each column.
- **D (Diagonal Detail):** The difference between the diagonal pixel values.

These four coefficients are computed for each 2×2 sub-block and serve as the features for further processing.

4.6 Feature Extraction

Feature extraction is a crucial step in image analysis, aiming to reduce the image's dimensionality while preserving important structural information. In the context of this approach, each 4×4 image block is first divided into four 2×2 sub-blocks. For each sub-block, the four Haar wavelet coefficients—Approximation (A), Horizontal

(H), Vertical (V), and Diagonal (D)—are computed. These coefficients effectively capture both the coarse and detailed components of the image, allowing for a more comprehensive representation of texture and edge information.

The approximation coefficient captures the average intensity, while the horizontal, vertical, and diagonal detail coefficients reflect intensity changes in different orientations. This multi-directional information is vital for tasks such as pattern recognition, face detection, and object tracking. Since each 2×2 sub-block contributes four coefficients, a single 4×4 block yields a total of 16 feature values. These extracted features form a compact yet informative feature set, enabling efficient classification or recognition when passed to subsequent stages of a machine learning or deep learning pipeline.

4.7 Feature Matrix Construction

The final feature matrix is constructed by aggregating the features extracted from all 4×4 blocks across the entire image. Each block contributes a set of 16 features, derived from the Haar wavelet coefficients of its four 2×2 sub-blocks. As this process is applied uniformly throughout the image, it ensures that both global and local variations in texture, intensity, and edge orientation are captured in a structured format. The aggregation of these feature sets across all blocks results in a high-dimensional matrix that provides a comprehensive representation of the image's content.

This feature matrix transforms the spatial domain information into a feature space where patterns and relationships among image regions become more explicit and distinguishable. Such a representation is especially useful in machine learning classification tasks, as it enables classifiers to operate on meaningful numerical attributes rather than raw pixel intensities. Additionally, the matrix format allows for the application of dimensionality reduction techniques (e.g., PCA, LDA) or normalization methods to enhance model performance. Thus, the construction of the feature matrix not only consolidates critical image information but also prepares the data for robust and efficient classification in tasks such as object recognition, face detection, or medical image analysis.

5 Results

This section presents the results of the block-wise transformation selection strategy applied to the input grayscale facial image. The image was divided into non-overlapping 4×4 blocks, and each block was subjected to four different Haar wavelet-based transformations: **O**, **T**, **L**, and **I**. Each of these transformations emphasizes different structural directions—diagonal, horizontal, vertical, and integrated features respectively—resulting in varied block-wise outputs. To identify the most discriminative transformation for each block, the L1 norm, computed as the sum of absolute pixel differences within the transformed block, was calculated for all four variants. The transformation that yielded the highest L1 norm was deemed the most significant in highlighting local variations and was thus selected to represent that block in the final feature map. This adaptive selection ensures that each region of the image

is encoded with the transformation that best captures its inherent texture, leading to improved representation for recognition tasks.

5.1 Block-wise Transformations and Visualization

In this section, we present and analyze the visual outcomes of applying four distinct transformations—**O**, **L**, **T**, and **I**—to each 4×4 block of the input grayscale facial image. These transformations are based on variations of the Haar wavelet technique, each emphasizing different structural features such as diagonals, verticals, horizontals, or combinations thereof. The application of these transforms alters the pixel arrangements within each block, highlighting specific directional patterns and localized textures. The transformed images not only showcase the impact of individual operations on facial structures but also provide visual cues for understanding the enhanced feature representations. For each transformation, a corresponding 16×16 block-level visualization is also provided, illustrating how local texture variations propagate across the entire image grid. These visual results demonstrate the ability of wavelet-based transformations to capture nuanced facial traits that are critical for accurate recognition.

5.1.1 Transformation O Results

Transformation **O** involves the application of a basic Haar wavelet transformation across non-overlapping 4×4 blocks of the grayscale facial image. This transformation aims to decompose local pixel intensities into frequency sub-bands, thereby capturing both coarse and fine structural information within each block. Notably, the Haar transform in this context emphasizes contrast variations along the diagonals, resulting in visible alterations that are most pronounced in the diagonal regions of the transformed image. This behavior is due to the diagonal detail coefficients (D) extracted during the Haar process, which are sensitive to diagonal edge patterns and intensity transitions. As a result, the block-wise transformation highlights subtle changes in facial contours, textures, and edge orientations, which are critical for distinguishing between different faces. The figure below illustrates the visual effect of Transformation **O**, where the diagonal emphasis becomes more evident, providing useful features for subsequent similarity comparisons and classification tasks. The effects of this transformation are shown in the figure below.

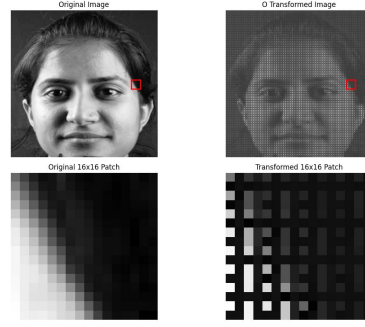


Fig. 3: Block-wise transformation using O (Original Transformation) applied to the image.

5.1.2 Transformation L Results

Transformation L applies a specialized variation of the wavelet transform to each 4×4 block, with an emphasis on extracting vertical detail coefficients from the Haar decomposition. Unlike Transformation O , which enhances diagonal features, Transformation L concentrates on capturing vertical intensity gradients by analyzing variations along the columns of each block. This transformation highlights vertical edges and line structures, making it particularly effective in emphasizing facial features such as the nose bridge, lips, or the outline of the face, which often exhibit strong vertical characteristics. The transformed output reflects these enhancements by producing contrast in vertical regions, which supports improved feature discrimination during the classification process. The figure below demonstrates the visual effect of Transformation L , where vertical details are amplified to create a clearer structural representation of the face. The effects of this transformation are shown in the figure below.

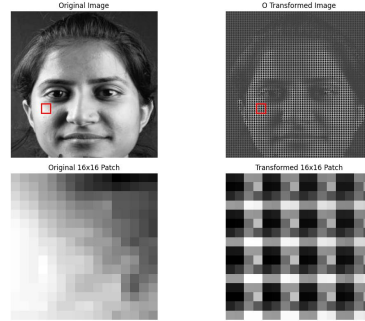


Fig. 4: L Transformation Results

5.1.3 Transformation T Results

Transformation **T** is designed to simultaneously incorporate both horizontal and vertical detail components derived from the Haar wavelet transform, applied to each 4×4 block of the grayscale face image. This dual modification approach enables the model to capture a richer set of features by emphasizing transitions along both axes. As a result, the transformation enhances the visibility of structural patterns such as facial outlines, eye and brow contours, and lip curvature — features that exhibit both horizontal and vertical variations. By combining these directional coefficients, Transformation **T** introduces increased complexity in the block-wise representation, yielding a texture-rich output that is particularly useful in differentiating subtle facial differences. The figure below illustrates how this transformation generates intricate patterns, reflecting multi-directional facial information critical for accurate face matching. The effects of this transformation are shown in the figure below.

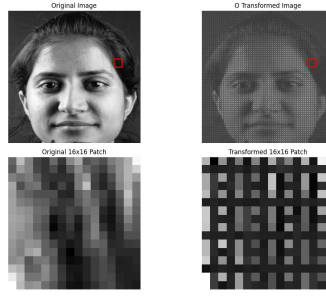


Fig. 5: T Transformation Results.

5.1.4 Transformation I Results

Transformation **I** integrates both horizontal and vertical Haar wavelet components in a comprehensive manner, resulting in the most detailed and complex transformation among all the variants. Applied on each 4×4 block of the grayscale facial image, this transformation extracts not only the basic approximation coefficients but also blends the vertical (V) and horizontal (H) details to form a densely encoded representation. The fusion of these directional features captures fine-grained facial textures, symmetrical features, and edge orientations that are critical for distinguishing between individuals, especially in scenarios involving subtle variations. The increased intricacy of the block-wise transformation allows for better retention of localized patterns while enhancing sensitivity to both axis-aligned and oblique structures. As a result, Transformation **I** produces a highly informative feature map that significantly boosts the discriminative power of the overall system. The figure below visually reflects the complexity introduced by this transformation across facial regions. This transformation is shown below.

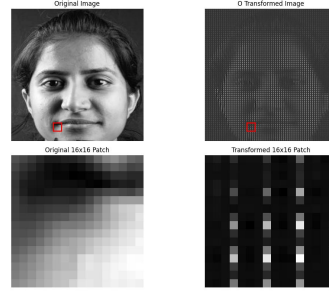


Fig. 6: I Transformation Results

These transformations demonstrate how different Haar wavelet techniques can influence the visual structure of the image at the block level, highlighting different coefficients and areas of change.

5.2 Transformation for Each Block

The transformation selected for each 4x4 block is shown below. The matrix represents the transformation applied to each block in the image:

$$\begin{bmatrix} L & L & L & L & \dots & L & L & L \\ L & L & L & L & \dots & L & L & L \\ L & L & L & L & \dots & L & L & L \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ L & L & L & L & \dots & O & L & L \\ L & L & L & L & \dots & I & L & L \end{bmatrix}$$

Each entry in the transformation matrix corresponds to the best transformation selected for the respective 4x4 block.

5.3 Prediction Results

This work presents a face recognition pipeline based on facial similarity analysis using MediaPipe, grayscale transformations, and Structural Similarity Index Measure (SSIM). The objective is to predict whether two facial images belong to the same person.

Face Extraction:

The pipeline initiates with the identification and extraction of facial regions from the input images using the **FaceDetection** module available in the MediaPipe library. Each image is initially converted from its original BGR format to the RGB color space, as required by MediaPipe's internal processing. Once converted, the face detection module locates facial landmarks and computes a bounding box that encapsulates the facial area. This bounding box is used to crop the face from the image, isolating it from the background and other non-relevant elements. The cropped face is subsequently

transformed into a grayscale image through a weighted combination of RGB channel values, preserving luminance information essential for structural analysis. Finally, the grayscale face is resized to a standardized resolution of 256×256 pixels, ensuring uniformity across the dataset and enabling consistent feature extraction and comparison in the subsequent stages of the recognition system.

Image Transformation:

Once the grayscale facial image has been standardized to a fixed resolution, it undergoes a sequence of four carefully designed custom transformations: **O**, **L**, **I**, and **T**. Each transformation is tailored to extract distinct structural or textural characteristics present in the facial region. Specifically, Transformation O captures diagonal features using basic Haar decomposition, L focuses on vertical details, T emphasizes both vertical and horizontal gradients with increased pattern complexity, and I integrates all components to provide a highly intricate representation. These transformations help enhance subtle differences between faces by exposing varied directional textures. The resulting images, including the original grayscale version and its four transformed variants, collectively form a comprehensive multi-view feature map. This enriched feature set provides diverse representations of the same face, improving the robustness and reliability of the similarity comparison process used in face recognition.

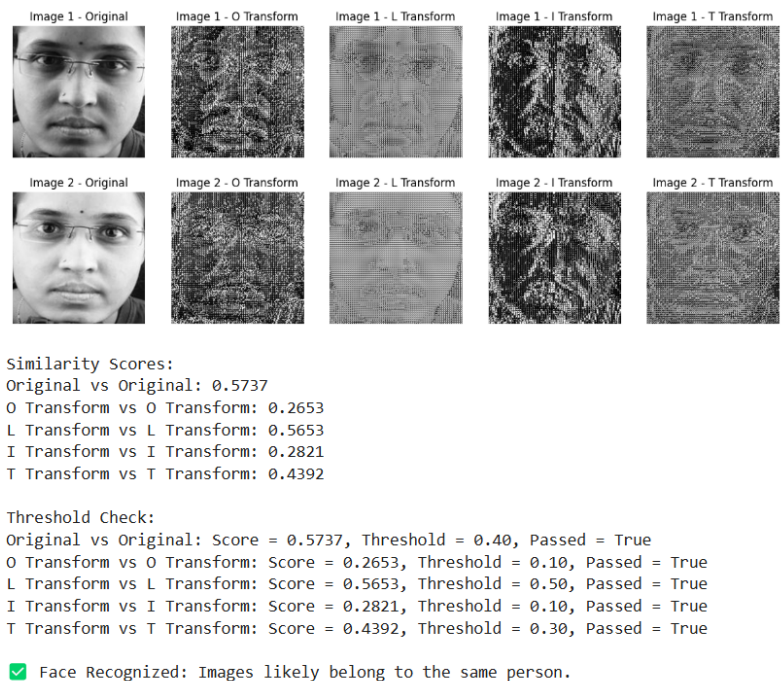
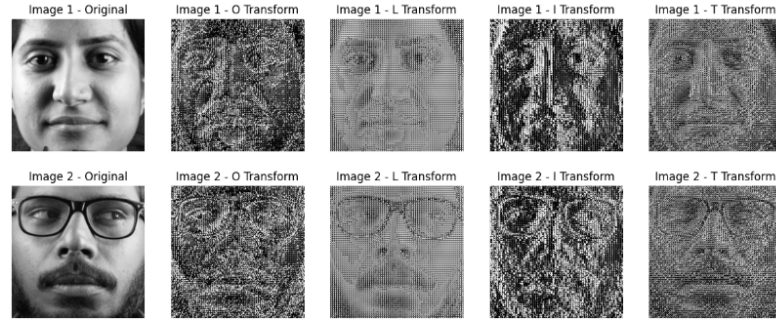


Fig. 7: Prediction - Face Match



Similarity Scores:
Original vs Original: 0.2205
O Transform vs O Transform: 0.0736
L Transform vs L Transform: 0.4948
I Transform vs I Transform: 0.0775
T Transform vs T Transform: 0.2679

Threshold Check:
Original vs Original: Score = 0.2205, Threshold = 0.40, Passed = False
O Transform vs O Transform: Score = 0.0736, Threshold = 0.10, Passed = False
L Transform vs L Transform: Score = 0.4948, Threshold = 0.50, Passed = False
I Transform vs I Transform: Score = 0.0775, Threshold = 0.10, Passed = False
T Transform vs T Transform: Score = 0.2679, Threshold = 0.30, Passed = False

❌ Face does NOT Match: Images likely belong to different people.

Fig. 8: Prediction - Face does not match

Similarity Measurement:

To compare the corresponding transformations of the two input faces, SSIM is computed for the following image pairs:

- Original vs Original
- O Transform vs O Transform
- L Transform vs L Transform
- I Transform vs I Transform
- T Transform vs T Transform

SSIM is a perceptual metric that measures the similarity between two images based on luminance, contrast, and structure. A score closer to 1 implies higher similarity.

Threshold-Based Classification:

Each transformation pair, generated from the original and altered facial images, is evaluated using the Structural Similarity Index Measure (SSIM), which quantifies the visual similarity between two images based on luminance, contrast, and structure. For each pair, a predefined similarity threshold is established through prior experimentation or validation on sample data. When the computed SSIM value for a given transformation exceeds its corresponding threshold, that specific comparison is flagged

as a successful match, indicating a strong resemblance between the facial features under that transformation. This matching process is repeated for all five transformation pairs. To ensure robustness and reliability in decision-making, the system adopts a majority-vote mechanism—if at least four out of the five transformation pairs produce SSIM values above their respective thresholds, the two input images are considered to represent the same individual. Conversely, if fewer than four pairs meet the similarity criteria, the system classifies the faces as belonging to different individuals, reflecting dissimilar structural or textural patterns.

Visualization:

The prediction results are visualized using side-by-side plots of the transformed faces. Each row displays the original and transformed outputs for the respective image. A sample result is shown in Figure 8.

6 Evaluation Results

This section outlines the methodology used to evaluate facial similarity between image pairs, incorporating both traditional face recognition techniques and transformation-based methods. The evaluation begins with a dataset of image pairs, each associated with a ground truth label that determines whether the pair represents a match or not. The images were processed using two different face detection techniques:

- The first technique uses Haar Cascade face detection, where faces are detected using OpenCV’s Haar Cascade Classifier. The cropped faces are resized to a fixed resolution of 256×256 pixels.
- The second technique employs MediaPipe-based face detection, where the faces are extracted and converted into grayscale images. These images undergo four directional transformations: O, I, T, and L, which capture various features of the face from different angles.

6.1 Similarity Metrics

To quantify the similarity between the face pairs, two primary metrics are utilized:

1. **Structural Similarity Index (SSIM)**: SSIM measures the perceptual similarity between two images by comparing their luminance, contrast, and structure. The SSIM formula is given by:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (5)$$

Where: - μ_x, μ_y are the mean pixel intensities of images x and y , - σ_x^2, σ_y^2 are the variances of x and y , - σ_{xy} is the covariance between the two images, - C_1 and C_2 are small constants to stabilize the calculation when the denominator is small.

2. **Local Binary Pattern Histogram (LBPH)**: LBPH computes a binary pattern for each pixel in a face image and then generates a histogram of these patterns.

The similarity is computed by comparing the histograms of two images. The LBPH formula is given by:

$$LBPH(x, y) = \sum_{i=0}^{n-1} |p_i - q_i| \quad (6)$$

Where: - p_i and q_i are the pixel values at the i -th position in the training and testing images, respectively, - n is the total number of pixels in the local region used for comparison.

6.2 Prediction Strategy

For the MediaPipe-based method, similarity scores are computed for each transformation (O, I, T, L) using SSIM and LBPH metrics. A decision-making mechanism is applied to determine the final prediction:

- A predefined threshold is applied to each transformation's SSIM and LBPH score.
- If at least three of the four transformations meet the threshold, the image pair is classified as a match.

For Haar-based face detection, the evaluation thresholds are as follows:

- SSIM threshold: 0.4
- LBPH threshold: 0.6

Predictions are recorded and stored in a CSV file for further analysis.

The performance of each model is evaluated by comparing the predicted labels with the actual ground truth labels. Key evaluation metrics such as accuracy, precision, recall, and F1-score are calculated, alongside confusion matrices to assess the overall performance. The accuracy formula is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Where: - TP = True Positives, TN = True Negatives, - FP = False Positives, FN = False Negatives.

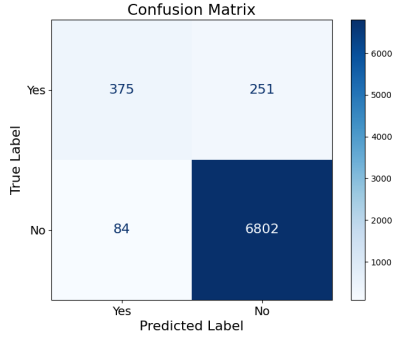
The results of the comparison reveal that the MediaPipe-based method, incorporating high-resolution image transformation and multi-directional comparison, outperforms the Haar-based method, which serves as a baseline.

6.3 Confusion Matrices and Results

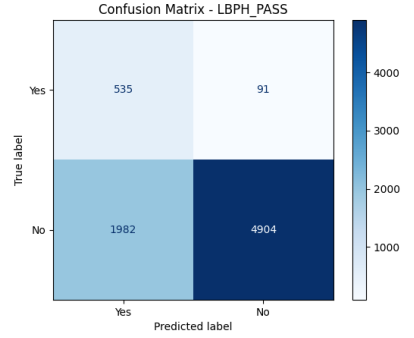
The confusion matrices and other evaluation metrics for both Haar and Tetrolet Transformation methods are presented below.

Method	SSIM Accuracy (%)	LBPH Accuracy (%)
Tetrolet Transformation	95.54	81.87
Haar Classifier	90.82	89.99

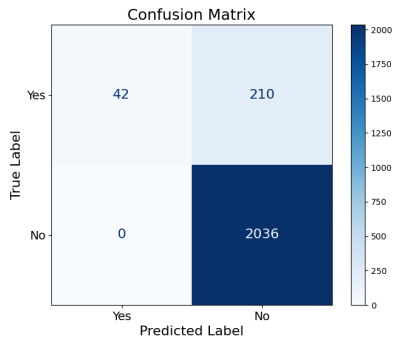
Table 1: Accuracy Comparison for Tetrolet and Haar Methods for SSIM and LBPH.



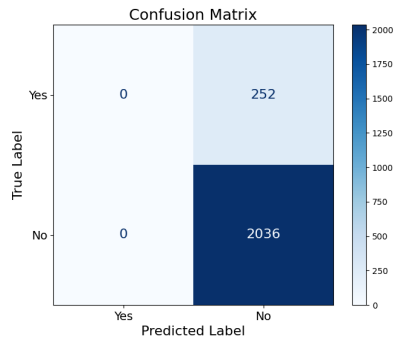
(a) Tetrolet SSIM Confusion Matrix



(b) Tetrolet LBPH Confusion Matrix



(c) Haar SSIM Confusion Matrix



(d) Haar LBPH Confusion Matrix

Fig. 9: Confusion matrices for SSIM and LBPH on Tetrolet and Haar-transformed images.

7 Conclusion

This project presents a detailed study on the use of wavelet-based neural networks for facial recognition, with a focus on integrating Tetrolet transforms to improve feature extraction. Unlike traditional convolutional neural networks (CNNs), which primarily operate on fixed-resolution pixel data, wavelet transforms allow for multiresolution analysis, capturing both spatial and frequency components of facial features. This

property makes the system highly effective in handling variations in lighting, facial orientation, and occlusion—common challenges in face recognition tasks. The Tetrolet transform, being a geometric adaptive variant of wavelets, further enhances the system by capturing local structures more accurately, leading to increased robustness and recognition precision.

By focusing on essential facial features while minimizing the influence of noise and irrelevant data, the system demonstrates strong consistency and reliability across varied scenarios. Its performance in real-time applications such as surveillance, identity verification, and access control highlights its practical value. Moreover, the architecture provides a scalable foundation that can be extended to more complex biometric systems. In summary, the fusion of wavelet techniques and deep learning offers a promising and powerful solution for next-generation facial recognition technologies, capable of delivering high accuracy and adaptability in dynamic environments.

The future of wavelet-based facial recognition systems presents many opportunities for advancement and deployment. A major development lies in extending the technology to 3D face recognition, which captures depth and structural details. This can enhance the system's accuracy even under challenging lighting or varied facial orientations. By adapting wavelet transforms to process volumetric data, the recognition system gains improved spatial representation and robustness across conditions.

Another direction is the integration of emotion and expression recognition. Tetrolet transforms, known for capturing local geometric features, can detect subtle facial cues and micro-expressions. This capability is valuable in behavioral analysis, health-care monitoring, and security. With emotional recognition, the system's functionality extends beyond identification, offering deeper insights in real-time scenarios.

Advancements also include deploying the model on edge platforms like mobile or embedded systems, ensuring real-time performance and privacy. Applications such as attendance systems, smart home security, and access control benefit from such integration. Combining facial recognition with biometrics like fingerprint or voice increases reliability. Incorporating federated learning and encrypted processing supports data protection compliance, making these systems more secure, ethical, and adoptable.

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Declarations

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- **Ethics approval and consent to participate:** Not applicable.
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- **Data availability:** The dataset used is publicly available on IEEE DataPort at: <https://ieee-dataport.org/documents/indian-facial-database-highlighting-spectacle-problems>.
- **Materials availability:** Not applicable.
- **Code availability:** The code used in this study is available from the corresponding author upon reasonable request.
- **Author contribution:** All authors contributed equally to the design, development, evaluation, and documentation of this work.

References

- [1] S. Mallat, “A Wavelet Tour of Signal Processing,” Academic Press, 1999.
- [2] L. T. Antony and S. N. Kumar, “Demystifying the Applications of Wavelet Transform in Bio Signal and Medical Data Processing,” in *Smart Trends in Computing and Communications. SmartCom 2024*, vol. 945, Springer, Singapore, 2024.
- [3] I. Daubechies, “The wavelet transform, time-frequency localization and signal analysis,” *IEEE Transactions on Information Theory*, vol. 36, no. 5, pp. 961–1005, 1990.
- [4] A. Jamal, M. A. Hameed Ashour, R. A. Abbas Helmi, and S. L. Fong, “A Wavelet-Neural Networks Model for Time Series,” *2021 IEEE 11th Symposium on Computer Applications & Industrial Electronics (ISCAIE)*, Penang, Malaysia, 2021, pp. 325–330.
- [5] M. Holschneider, “Wavelets: An Analysis Tool,” Springer, 1990.
- [6] M. Saha, M. K. Naskar, and B. N. Chatterji, “Advanced Wavelet Transform for Image Processing—A Survey,” in *Information, Photonics and Communication*, vol. 79, Springer, Singapore, 2020.
- [7] T. Guo, T. Zhang, E. Lim, M. López-Benítez, F. Ma, and L. Yu, “A review of wavelet analysis and its applications: Challenges and opportunities,” *IEEE Access*, vol. 10, pp. 58869–58903, 2022, doi: 10.1109/ACCESS.2022.3179517.
- [8] Y. Pati and P. Krishnaprasad, “Analysis and synthesis of feedforward neural networks using discrete affine wavelet transforms,” *IEEE Transactions on Neural Networks*, vol. 4, no. 1, pp. 73–85, 1993.
- [9] Oscar Herrera, Belém Priego, David Pinto, Beatriz Beltrán, and Vivek Singh. “Wavelets as activation functions in Neural Networks.” *Journal of Intelligent & Fuzzy Systems*, vol. 42, no. 5, 2022, pp. 4345–4355.
- [10] A. D. Le, S. Jin, Y. S. Bae and T. Nguyen, “A Novel Learnable Orthogonal Wavelet Unit Neural Network with Perfection Reconstruction Constraint Relaxation for Image Classification,” *2023 IEEE International Conference on Visual*

Communications and Image Processing (VCIP), Jeju, Korea, Republic of, 2023, pp. 1-5, doi: 10.1109/VCIP59821.2023.10402772.

- [11] A. K. Alexandridis and A. D. Zaprani, "Wavelet neural networks: A practical guide," *Neural Networks*, vol. 42, 2013, pp. 1–27. ISSN 0893-6080.
- [12] T. Wang, V. Guercio, P. Cattani, and F. Villecco, "A Review of Research Progress and Application of Wavelet Neural Networks," in *New Technologies, Development and Application VI. NT 2023*, vol. 687, Springer, Cham, 2023.
- [13] R. Xu, Y. Xu, X. Yang, H. Huang, Z. Lei, and Y. Quan, "Wavelet analysis model inspired convolutional neural networks for image denoising," *Applied Mathematical Modelling*, 2023.
- [14] Q. Zhang and A. Benveniste, "Wavelet networks," *IEEE Transactions on Neural Networks*, vol. 3, no. 6, pp. 889–898, 1992.
- [15] Szu, Harold, Telfer, Brian, and Kadambe, Shubha. "Neural network adaptive wavelets for signal representation and classification" *Optical Engineering*, vol. 31, 1992, pp. 1907–1916.
- [16] P. Liang, C. Deng, J. Wu, and Z. Yang, "Intelligent fault diagnosis of rotating machinery via wavelet transform, generative adversarial nets and convolutional neural network," *Measurement*, 2020, article 107768.
- [17] Ramzi, Z., Michalewicz, K., Starck, J. L., et al. "Wavelets in the Deep Learning Era." *Journal of Mathematical Imaging and Vision*, vol. 65, 2023, pp. 240–251.
- [18] Y. Zhao, Y. Shen, Y. Zhu, and J. Yao, "Forecasting Wavelet Transformed Time Series with Attentive Neural Networks," *2018 IEEE International Conference on Data Mining (ICDM)*, Singapore, 2018, pp. 1452–1457. doi: 10.1109/ICDM.2018.00201.
- [19] L. Mayaluri, S. Gupta, and N. Panda, "An Indian facial database highlighting the Spectacle Problems," *IEEE Dataport*, 2019.

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